Program Summary - Ridership_Regression_PostCovid.sas

Execution Environment

Author:	u64420076
File:	/home/u64420076/BAN000_Personal/Ridership_Regression_PostCovid.sas
SAS Platform:	Linux LIN X64 5.14.0-284.30.1.el9_2.x86_64
SAS Host:	ODAWS01-USW2.ODA.SAS.COM
SAS Version:	9.04.01M8P02222023
SAS Locale:	en_US
Submission Time:	4/17/2026, 2:35:26 PM
Browser Host:	POOL-173-35-226-51.CPE.NET.CABLE.ROGERS.COM
User Agent:	Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/147.0.0.0 Safari/537.36
Application Server:	ODAMID00-USW2.ODA.SAS.COM

Code: Ridership_Regression_PostCovid.sas

```
/* Importr Data */

FILENAME REFFILE '/home/u64420076/BAN000_Personal/main_dataset_modeling.csv';

PROC IMPORT DATAFILE=REFFILE
    DBMS=CSV
    OUT=WORK.TRANSIT;
    GETNAMES=YES;
RUN;

/* dummy variables */

data transit_reg;
    set transit;

    /* Month dummies: month 12 is reference */
    month_1  = (month = 1);
    month_2  = (month = 2);
    month_3  = (month = 3);
    month_4  = (month = 4);
    month_5  = (month = 5);
    month_6  = (month = 6);
    month_7  = (month = 7);
    month_8  = (month = 8);
    month_9  = (month = 9);
    month_10 = (month = 10);
    month_11 = (month = 11);

    /* COVID period dummies: Pre-COVID is reference */
    covidrec  = (covid_period = "COVID/Rec");
    postcovid = (covid_period = "Post-COVI");
run;

/* create period subsets */
data pre_covid2 ;
    set transit_reg;
    if covid_period = "Pre-COVID" then output pre_covid2;
run;

data covid_rec2;
    set transit_reg;
    if covid_period = "COVID/Rec" then output covid_rec2;
run;

data post_covid2;
    set transit_reg;
    if covid_period = "Post-COVI" then output post_covid2;
run;

/*_____________________________________POST-COVID TEST MODELS_____________________________________*/

/* covid_rec2 Time Baseline Model */
proc reg data=post_covid2;
    model ridership =
        month_1 month_2 month_3 month_4 month_5 month_6
        month_7 month_8 month_9 month_10 month_11

        / vif tol;
```

```
    title "PROC REG Model 1: Baseline";
run; quit;

/* covid_rec2 Lag Baseline Model */
proc reg data=post_covid2;
    model ridership =
        month_1 month_2 month_3 month_4 month_5 month_6
        month_7 month_8 month_9 month_10 month_11
        ridership_lag1 ridership_lag12
        / vif tol;
    title "PROC REG Model 2: Lag Baseline";
run; quit;

/* covid_rec2 full model */
proc reg data=post_covid2;
    model ridership=
        month_1 month_2 month_3 month_4 month_5 month_6
        month_7 month_8 month_9 month_10 month_11
        ridership_lag1 ridership_lag12
        gas_price_lag1 unemployment_rate_lag1
        avg_temp total_precip_mm total_snow_cm
        / vif tol ;
     title 'Pre-Covid PROC REG Model 3.1';

run; quit;

/* step-wise */
proc reg data=post_covid2;
    model ridership=
        month_1 month_2 month_3 month_4 month_5 month_6
        month_7 month_8 month_9 month_10 month_11
        covidrec postcovid
        ridership_lag1 ridership_lag12
        gas_price_lag1 unemployment_rate_lag1
        avg_temp total_precip_mm total_snow_cm
        / vif tol SELECTION=STEPWISE SLENTRY=0.05 SLSTAY=0.15;
     title 'Pre-Covid PROC REG STEP Model 3.2';

run; quit;

/*_____________________________________POST-COVID MODELS_____________________________________*/


/* Month-only Seasonal Baseline */
proc reg data=post_covid2;
    model ridership =
        month_1 month_2 month_3 month_4 month_5 month_6
        month_7 month_8 month_9 month_10 month_11
        / vif tol;
    title "Pre-COVID Model P1: Month-only Seasonal Baseline";
run; quit;


/* Lag Memory */
proc reg data=post_covid2;
    model ridership =
        ridership_lag1
        ridership_lag12
        / vif tol;
    title "Pre-COVID Model P2: Lag Memory Model";
run; quit;

/* Reduced External Model */
proc reg data=post_covid2;
    model ridership =
        ridership_lag1
        ridership_lag12
        gas_price_lag1
        unemployment_rate_lag1
        / vif tol;
    title "Pre-COVID Model P3: Reduced External Model";
run; quit;



proc reg data=post_covid2;
    model ridership=
        gas_price_lag1 unemployment_rate_lag1
        avg_temp total_precip_mm total_snow_cm
```

```
         / vif tol ;
      title 'Pre-Covid Model P4: Whithout Ridership History';

  run; quit;
```

Log: Ridership_Regression_PostCovid.sas

Notes (27)

```
1          OPTIONS NONOTES NOSTIMER NOSOURCE NOSYNTAXCHECK;
68
69         /* Importr Data */
70
71         FILENAME REFFILE '/home/u64420076/BAN000_Personal/main_dataset_modeling.csv';
72
73         PROC IMPORT DATAFILE=REFFILE
74         DBMS=CSV
75         OUT=WORK.TRANSIT;
76         GETNAMES=YES;
77         RUN;

NOTE: Unable to open parameter catalog: SASUSER.PARMS.PARMS.SLIST in update mode. Temporary parameter values will be saved to
WORK.PARMS.PARMS.SLIST.
78          /*********************************************************************
79          *   PRODUCT:   SAS
80          *   VERSION:   9.4
81          *   CREATOR:   External File Interface
82          *   DATE:      17APR26
83          *   DESC:      Generated SAS Datastep Code
84          *   TEMPLATE SOURCE:  (None Specified.)
85          *********************************************************************/
86             data WORK.TRANSIT    ;
87             %let _EFIERR_ = 0; /* set the ERROR detection macro variable */
88             infile REFFILE delimiter = ',' MISSOVER DSD  firstobs=2 ;
89                info format "Unnamed: 0"N best32. ;
90                info format date yymmdd10. ;
91                info format year best32. ;
92                info format month best32. ;
93                info format quarter best32. ;
94                info format season $6. ;
95                info format covid_period $9. ;
96                info format ridership best32. ;
97                info format gas_price_regular best32. ;
98                info format unemployment_rate best32. ;
99                info format avg_temp best32. ;
100               info format total_precip_mm best32. ;
101               info format total_snow_cm best32. ;
102               info format avg_max_temp best32. ;
103               info format avg_min_temp best32. ;
104               info format climate_data_flag $5. ;
105               info format days_in_month best32. ;
106               info format missing_temp_days best32. ;
107               info format missing_precip_days best32. ;
108               info format missing_snow_days best32. ;
109               info format temp_completeness best32. ;
110               info format precip_completeness best32. ;
111               info format snow_completeness best32. ;
112               info format ridership_lag1 best32. ;
113               info format ridership_lag12 best32. ;
```

```
114             informat gas_price_lag1 best32. ;
115             informat unemployment_rate_lag1 best32. ;
116             informat avg_temp_lag1 best32. ;
117             informat total_precip_mm_lag1 best32. ;
118             informat total_snow_cm_lag1 best32. ;
119             format "Unnamed: 0"N best12. ;
120             format date yymmdd10. ;
121             format year best12. ;
122             format month best12. ;
123             format quarter best12. ;
124             format season $6. ;
125             format covid_period $9. ;
126             format ridership best12. ;
127             format gas_price_regular best12. ;
128             format unemployment_rate best12. ;
129             format avg_temp best12. ;
130             format total_precip_mm best12. ;
131             format total_snow_cm best12. ;
132             format avg_max_temp best12. ;
133             format avg_min_temp best12. ;
134             format climate_data_flag $5. ;
135             format days_in_month best12. ;
136             format missing_temp_days best12. ;
137             format missing_precip_days best12. ;
138             format missing_snow_days best12. ;
139             format temp_completeness best12. ;
140             format precip_completeness best12. ;
141             format snow_completeness best12. ;
142             format ridership_lag1 best12. ;
143             format ridership_lag12 best12. ;
144             format gas_price_lag1 best12. ;
145             format unemployment_rate_lag1 best12. ;
146             format avg_temp_lag1 best12. ;
147             format total_precip_mm_lag1 best12. ;
148             format total_snow_cm_lag1 best12. ;
149          input
150                      "Unnamed: 0"N
151                      date
152                      year
153                      month
154                      quarter
155                      season  $
156                      covid_period  $
157                      ridership
158                      gas_price_regular
159                      unemployment_rate
160                      avg_temp
161                      total_precip_mm
162                      total_snow_cm
163                      avg_max_temp
164                      avg_min_temp
165                      climate_data_flag  $
166                      days_in_month
167                      missing_temp_days
168                      missing_precip_days
169                      missing_snow_days
170                      temp_completeness
171                      precip_completeness
172                      snow_completeness
173                      ridership_lag1
174                      ridership_lag12
175                      gas_price_lag1
176                      unemployment_rate_lag1
177                      avg_temp_lag1
178                      total_precip_mm_lag1
179                      total_snow_cm_lag1
180          ;
181          if _ERROR_ then call symputx('_EFIERR_',1);  /* set ERROR detection macro variable */
182          run;

NOTE: The infile REFFILE is:
      Filename=/home/u64420076/BAN000_Personal/main_dataset_modeling.csv,
      Owner Name=u64420076,Group Name=oda,
      Access Permission=-rw-r--r--,
      Last Modified=16Apr2026:13:26:50,
      File Size (bytes)=38716

NOTE: 179 records were read from the infile REFFILE.
      The minimum record length was 166.
      The maximum record length was 267.
NOTE: The data set WORK.TRANSIT has 179 observations and 30 variables.
NOTE: DATA statement used (Total process time):
      real time           0.00 seconds
      user cpu time       0.01 seconds
      system cpu time     0.00 seconds
      memory              10804.93k
      OS Memory           30240.00k
      Timestamp           04/17/2026 06:35:21 PM
      Step Count                        37  Switch Count  2
```

```
      Page Faults                       0
      Page Reclaims                     209
      Page Swaps                        0
      Voluntary Context Switches        16
      Involuntary Context Switches      0
      Block Input Operations            0
      Block Output Operations           272


179 rows created in WORK.TRANSIT from REFFILE.



NOTE: WORK.TRANSIT data set was successfully created.
NOTE: The data set WORK.TRANSIT has 179 observations and 30 variables.
NOTE: PROCEDURE IMPORT used (Total process time):
      real time           0.06 seconds
      user cpu time       0.04 seconds
      system cpu time     0.01 seconds
      memory              10804.93k
      OS Memory           30500.00k
      Timestamp           04/17/2026 06:35:21 PM
      Step Count                        37  Switch Count  8
      Page Faults                       0
      Page Reclaims                     2900
      Page Swaps                        0
      Voluntary Context Switches        99
      Involuntary Context Switches      2
      Block Input Operations            0
      Block Output Operations           368


183
184       /* dummy variables */
185
186       data transit_reg;
187           set transit;
188
189           /* Month dummies: month 12 is reference */
190           month_1  = (month = 1);
191           month_2  = (month = 2);
192           month_3  = (month = 3);
193           month_4  = (month = 4);
194           month_5  = (month = 5);
195           month_6  = (month = 6);
196           month_7  = (month = 7);
197           month_8  = (month = 8);
198           month_9  = (month = 9);
199           month_10 = (month = 10);
200           month_11 = (month = 11);
201
202           /* COVID period dummies: Pre-COVID is reference */
203           covidrec  = (covid_period = "COVID/Rec");
204           postcovid = (covid_period = "Post-COVI");
205       run;

NOTE: There were 179 observations read from the data set WORK.TRANSIT.
NOTE: The data set WORK.TRANSIT_REG has 179 observations and 43 variables.
NOTE: DATA statement used (Total process time):
      real time           0.00 seconds
      user cpu time       0.00 seconds
      system cpu time     0.00 seconds
      memory              1057.53k
      OS Memory           24752.00k
      Timestamp           04/17/2026 06:35:21 PM
      Step Count                        38  Switch Count  2
      Page Faults                       0
      Page Reclaims                     116
      Page Swaps                        0
      Voluntary Context Switches        10
      Involuntary Context Switches      0
      Block Input Operations            0
      Block Output Operations           264


206
207       /* create period subsets */
208       data pre_covid2 ;
209           set transit_reg;
210           if covid_period = "Pre-COVID" then output pre_covid2;
211       run;

NOTE: There were 179 observations read from the data set WORK.TRANSIT_REG.
NOTE: The data set WORK.PRE_COVID2 has 109 observations and 43 variables.
NOTE: DATA statement used (Total process time):
      real time           0.00 seconds
      user cpu time       0.00 seconds
      system cpu time     0.01 seconds
      memory              1055.71k
```

```
      OS Memory           24752.00k
      Timestamp           04/17/2026 06:35:21 PM
      Step Count                        39  Switch Count  2
      Page Faults                       0
      Page Reclaims                     112
      Page Swaps                        0
      Voluntary Context Switches        12
      Involuntary Context Switches      0
      Block Input Operations            0
      Block Output Operations           264


212
213        data covid_rec2;
214            set transit_reg;
215            if covid_period = "COVID/Rec" then output covid_rec2;
216        run;

NOTE: There were 179 observations read from the data set WORK.TRANSIT_REG.
NOTE: The data set WORK.COVID_REC2 has 34 observations and 43 variables.
NOTE: DATA statement used (Total process time):
      real time           0.00 seconds
      user cpu time       0.01 seconds
      system cpu time     0.00 seconds
      memory              1055.84k
      OS Memory           24752.00k
      Timestamp           04/17/2026 06:35:21 PM
      Step Count                        40  Switch Count  2
      Page Faults                       0
      Page Reclaims                     117
      Page Swaps                        0
      Voluntary Context Switches        10
      Involuntary Context Switches      0
      Block Input Operations            0
      Block Output Operations           264


217
218        data post_covid2;
219            set transit_reg;
220            if covid_period = "Post-COVI" then output post_covid2;
221        run;

NOTE: There were 179 observations read from the data set WORK.TRANSIT_REG.
NOTE: The data set WORK.POST_COVID2 has 36 observations and 43 variables.
NOTE: DATA statement used (Total process time):
      real time           0.00 seconds
      user cpu time       0.00 seconds
      system cpu time     0.00 seconds
      memory              1055.71k
      OS Memory           24752.00k
      Timestamp           04/17/2026 06:35:21 PM
      Step Count                        41  Switch Count  2
      Page Faults                       0
      Page Reclaims                     117
      Page Swaps                        0
      Voluntary Context Switches        10
      Involuntary Context Switches      0
      Block Input Operations            0
      Block Output Operations           264


222
223        /*_________________________________________POST-COVID TEST MODELS_________________________________________*/
224
225        /* covid_rec2 Time Baseline Model */
226        proc reg data=post_covid2;
227            model ridership =
228                month_1 month_2 month_3 month_4 month_5 month_6
229                month_7 month_8 month_9 month_10 month_11
230
231                / vif tol;
232            title "PROC REG Model 1: Baseline";
233        run;

233      !      quit;

NOTE: PROCEDURE REG used (Total process time):
      real time           2.84 seconds
      user cpu time       0.17 seconds
      system cpu time     0.04 seconds
      memory              14674.68k
      OS Memory           37708.00k
      Timestamp           04/17/2026 06:35:24 PM
      Step Count                        42  Switch Count  23
      Page Faults                       0
      Page Reclaims                     15514
      Page Swaps                        0
      Voluntary Context Switches        1033
```

```
      Involuntary Context Switches      8
      Block Input Operations            0
      Block Output Operations           1192


234
235        /* covid_rec2 Lag Baseline Model */
236        proc reg data=post_covid2;
237            model ridership =
238                month_1 month_2 month_3 month_4 month_5 month_6
239                month_7 month_8 month_9 month_10 month_11
240                ridership_lag1 ridership_lag12
241                / vif tol;
242            title "PROC REG Model 2: Lag Baseline";
243        run;

243      !      quit;

NOTE: PROCEDURE REG used (Total process time):
      real time           0.56 seconds
      user cpu time       0.19 seconds
      system cpu time     0.03 seconds
      memory              11451.84k
      OS Memory           43496.00k
      Timestamp           04/17/2026 06:35:24 PM
      Step Count                        43  Switch Count  23
      Page Faults                       0
      Page Reclaims                     12613
      Page Swaps                        0
      Voluntary Context Switches        1002
      Involuntary Context Switches      15
      Block Input Operations            0
      Block Output Operations           1432


244
245        /* covid_rec2 full model */
246        proc reg data=post_covid2;
247        model ridership=
248        month_1 month_2 month_3 month_4 month_5 month_6
249                month_7 month_8 month_9 month_10 month_11
250                ridership_lag1 ridership_lag12
251                gas_price_lag1 unemployment_rate_lag1
252                avg_temp total_precip_mm total_snow_cm
253                / vif tol ;
254             title 'Pre-Covid PROC REG Model 3.1';
255
256        run;

256      !      quit;

NOTE: PROCEDURE REG used (Total process time):
      real time           0.56 seconds
      user cpu time       0.22 seconds
      system cpu time     0.03 seconds
      memory              11263.84k
      OS Memory           43240.00k
      Timestamp           04/17/2026 06:35:25 PM
      Step Count                        44  Switch Count  23
      Page Faults                       0
      Page Reclaims                     12466
      Page Swaps                        0
      Voluntary Context Switches        1064
      Involuntary Context Switches      13
      Block Input Operations            0
      Block Output Operations           1472


257
258        /* step-wise */
259        proc reg data=post_covid2;
260        model ridership=
261        month_1 month_2 month_3 month_4 month_5 month_6
262                month_7 month_8 month_9 month_10 month_11
263                covidrec postcovid
264                ridership_lag1 ridership_lag12
265                gas_price_lag1 unemployment_rate_lag1
266                avg_temp total_precip_mm total_snow_cm
267                / vif tol SELECTION=STEPWISE SLENTRY=0.05 SLSTAY=0.15;
268             title 'Pre-Covid PROC REG STEP Model 3.2';
269
270        run;

270      !      quit;

NOTE: PROCEDURE REG used (Total process time):
      real time           0.35 seconds
      user cpu time       0.17 seconds
      system cpu time     0.03 seconds
```

```
memory              11268.81k
OS Memory           42984.00k
Timestamp           04/17/2026 06:35:25 PM
Step Count                        45  Switch Count  23
Page Faults                       0
Page Reclaims                     12037
Page Swaps                        0
Voluntary Context Switches        688
Involuntary Context Switches      4
Block Input Operations            0
Block Output Operations           736


271
272        /*_____________________________________POST-COVID MODELS_____________________________________*/
273
274
275        /* Month-only Seasonal Baseline */
276        proc reg data=post_covid2;
277            model ridership =
278                month_1 month_2 month_3 month_4 month_5 month_6
279                month_7 month_8 month_9 month_10 month_11
280                / vif tol;
281            title "Pre-COVID Model P1: Month-only Seasonal Baseline";
282        run;

282      !      quit;

NOTE: PROCEDURE REG used (Total process time):
      real time           0.42 seconds
      user cpu time       0.16 seconds
      system cpu time     0.02 seconds
      memory              11201.56k
      OS Memory           43240.00k
      Timestamp           04/17/2026 06:35:26 PM
      Step Count                        46  Switch Count  23
      Page Faults                       0
      Page Reclaims                     12247
      Page Swaps                        0
      Voluntary Context Switches        870
      Involuntary Context Switches      9
      Block Input Operations            0
      Block Output Operations           896


283
284
285        /* Lag Memory */
286        proc reg data=post_covid2;
287            model ridership =
288                ridership_lag1
289                ridership_lag12
290                / vif tol;
291            title "Pre-COVID Model P2: Lag Memory Model";
292        run;

292      !      quit;

NOTE: PROCEDURE REG used (Total process time):
      real time           0.25 seconds
      user cpu time       0.12 seconds
      system cpu time     0.03 seconds
      memory              11242.31k
      OS Memory           43240.00k
      Timestamp           04/17/2026 06:35:26 PM
      Step Count                        47  Switch Count  23
      Page Faults                       0
      Page Reclaims                     11968
      Page Swaps                        0
      Voluntary Context Switches        688
      Involuntary Context Switches      6
      Block Input Operations            0
      Block Output Operations           720


293
294        /* Reduced External Model */
295        proc reg data=post_covid2;
296            model ridership =
297                ridership_lag1
298                ridership_lag12
299                gas_price_lag1
300                unemployment_rate_lag1
301                / vif tol;
302            title "Pre-COVID Model P3: Reduced External Model";
303        run;

303      !      quit;
```

```
NOTE: PROCEDURE REG used (Total process time):
      real time           0.29 seconds
      user cpu time       0.14 seconds
      system cpu time     0.03 seconds
      memory              11157.18k
      OS Memory           43240.00k
      Timestamp           04/17/2026 06:35:26 PM
      Step Count                        48  Switch Count  23
      Page Faults                       0
      Page Reclaims                     12080
      Page Swaps                        0
      Voluntary Context Switches        699
      Involuntary Context Switches      4
      Block Input Operations            0
      Block Output Operations           776


304
305
306
307        proc reg data=post_covid2;
308        model ridership=
309                gas_price_lag1 unemployment_rate_lag1
310                avg_temp total_precip_mm total_snow_cm
311                / vif tol ;
312             title 'Pre-Covid Model P4: Whithout Ridership History';
313
314        run;
314      !      quit;

NOTE: PROCEDURE REG used (Total process time):
      real time           0.29 seconds
      user cpu time       0.12 seconds
      system cpu time     0.02 seconds
      memory              11209.59k
      OS Memory           43496.00k
      Timestamp           04/17/2026 06:35:27 PM
      Step Count                        49  Switch Count  23
      Page Faults                       0
      Page Reclaims                     12018
      Page Swaps                        0
      Voluntary Context Switches        717
      Involuntary Context Switches      8
      Block Input Operations            0
      Block Output Operations           768


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343        OPTIONS NONOTES NOSTIMER NOSOURCE NOSYNTAXCHECK;
353
```

Results: Ridership_Regression_PostCovid.sas

PROC REG Model 1: Baseline

The REG Procedure
Model: MODEL1
Dependent Variable: ridership

Number of Observations Read	36

Number of Observations Used	36

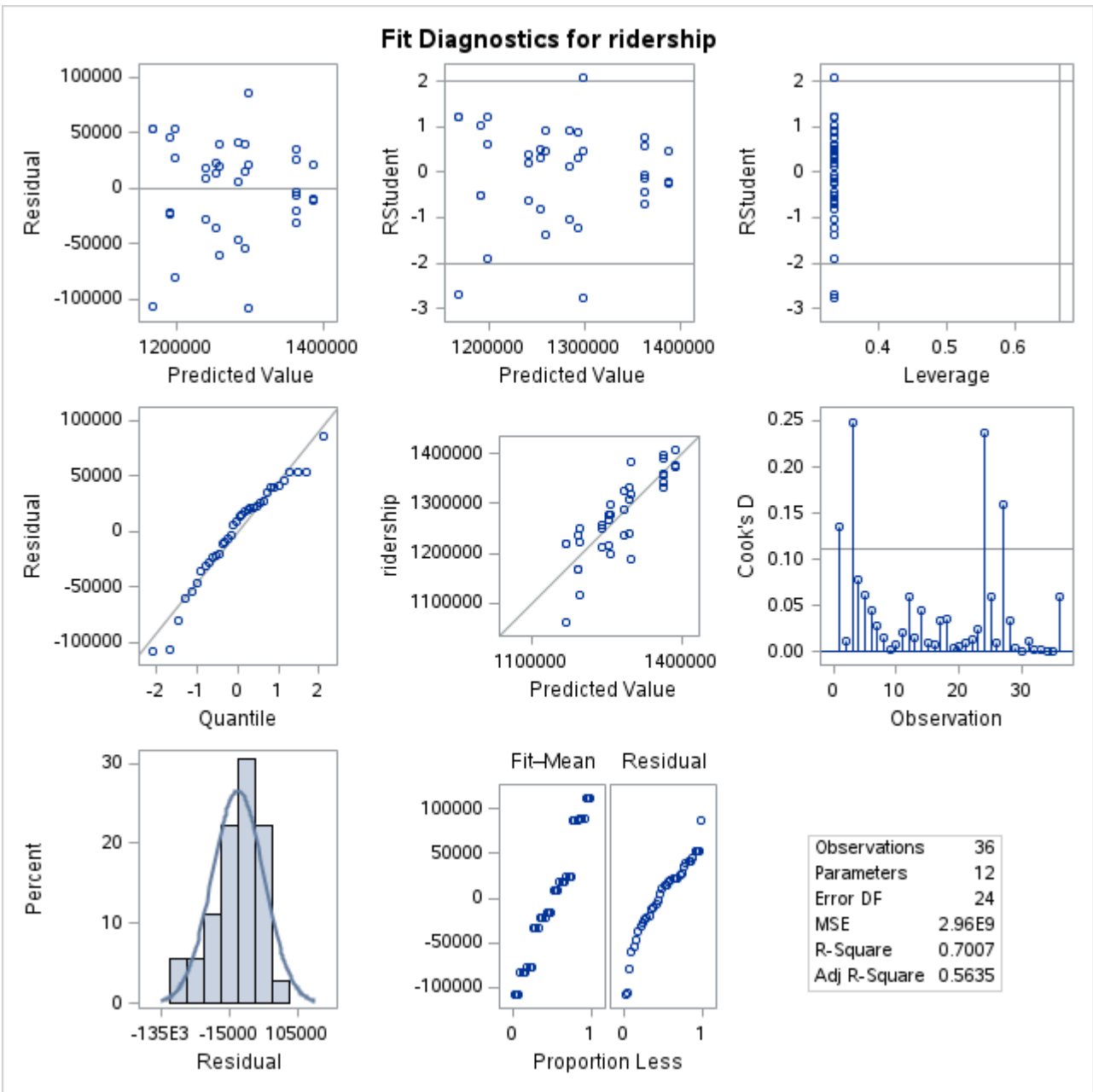
Analysis of Variance					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	11	1.664229E11	15129353535	5.11	0.0004
Error	24	71096000000	2962333333		
Corrected Total	35	2.375189E11			

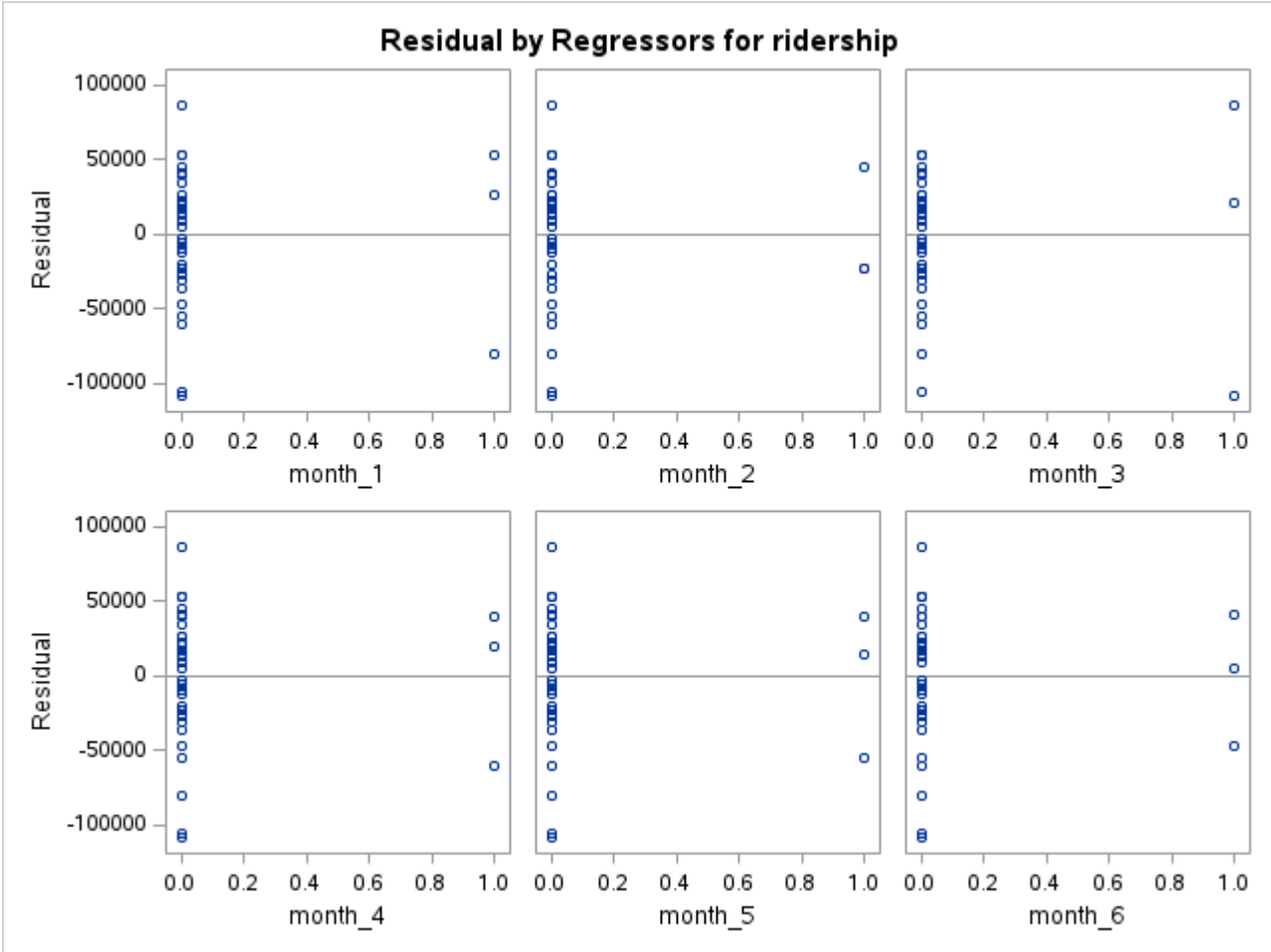
Root MSE	54427	R-Square	0.7007
Dependent Mean	1274556	Adj R-Sq	0.5635
Coeff Var	4.27030		

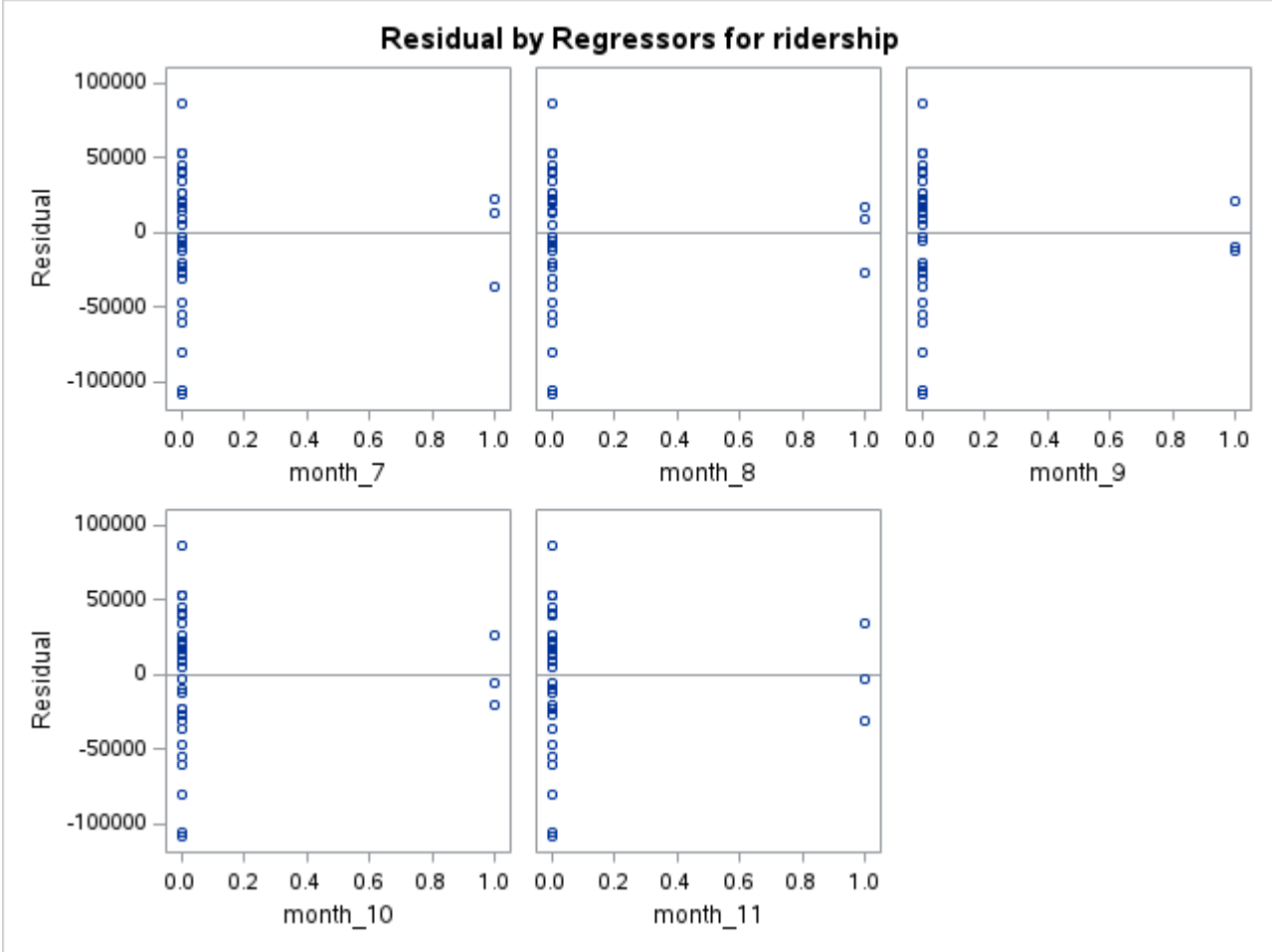
Parameter Estimates							
Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > |t|	Tolerance	Variance Inflation
Intercept	1	1167000	31424	37.14	<.0001	.	0
month_1	1	30000	44440	0.68	0.5061	0.54545	1.83333
month_2	1	24333	44440	0.55	0.5891	0.54545	1.83333
month_3	1	131333	44440	2.96	0.0069	0.54545	1.83333
month_4	1	91667	44440	2.06	0.0501	0.54545	1.83333
month_5	1	126333	44440	2.84	0.0090	0.54545	1.83333
month_6	1	116333	44440	2.62	0.0151	0.54545	1.83333
month_7	1	86333	44440	1.94	0.0639	0.54545	1.83333
month_8	1	73333	44440	1.65	0.1119	0.54545	1.83333
month_9	1	219667	44440	4.94	<.0001	0.54545	1.83333
month_10	1	196000	44440	4.41	0.0002	0.54545	1.83333
month_11	1	195333	44440	4.40	0.0002	0.54545	1.83333

PROC REG Model 1: Baseline

The REG Procedure
Model: MODEL1
Dependent Variable: ridership







PROC REG Model 2: Lag Baseline

The REG Procedure
Model: MODEL1
Dependent Variable: ridership

Number of Observations Read	36
Number of Observations Used	36

Analysis of Variance

Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	13	1.962954E11	15099644795	8.06	<.0001
Error	22	41223506551	1873795752		
Corrected Total	35	2.375189E11			

Root MSE	43287	**R-Square**	0.8264
Dependent Mean	1274556	**Adj R-Sq**	0.7239
Coeff Var	3.39627		

Parameter Estimates

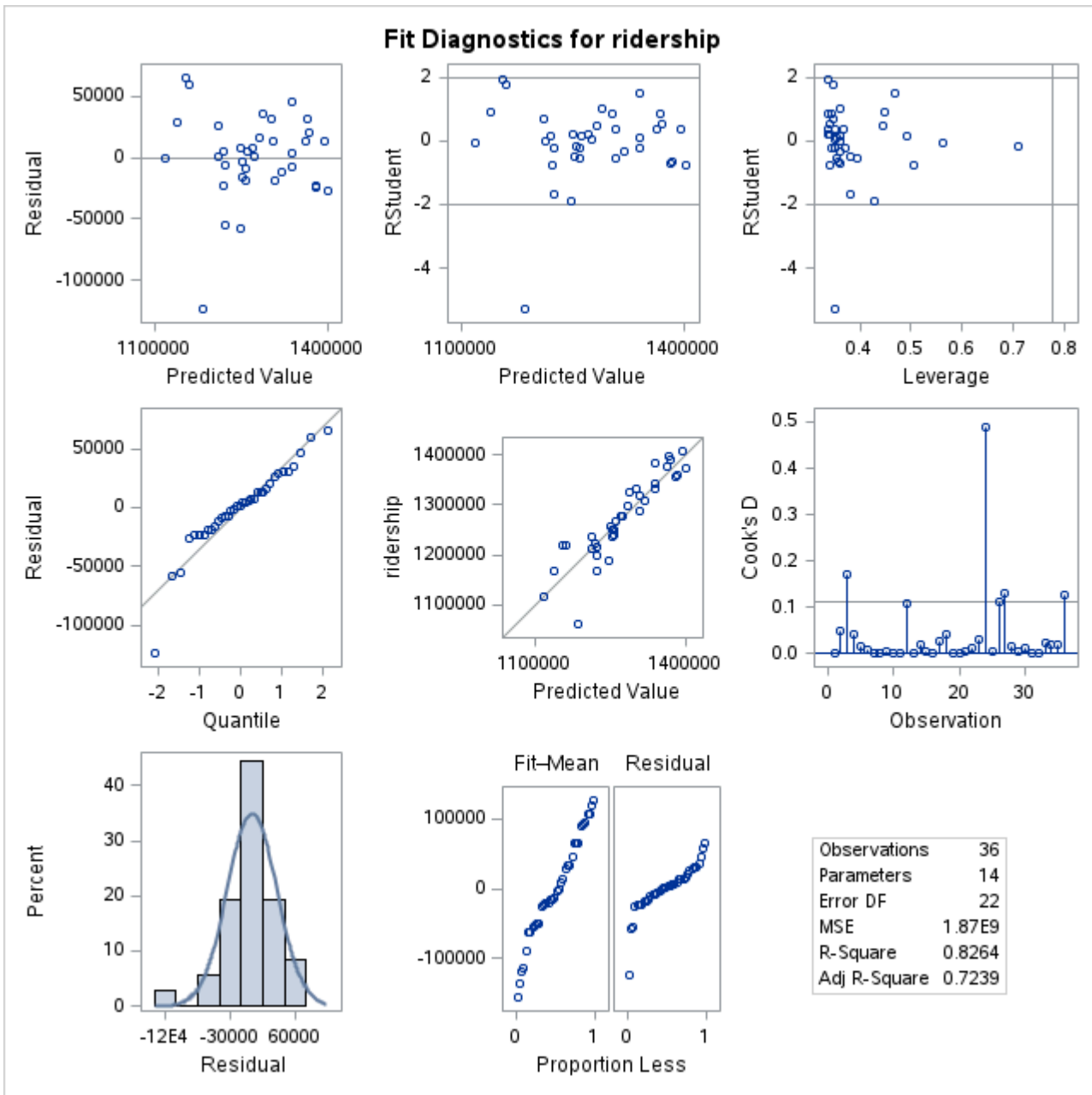
Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > |t|	Tolerance	Variance Inflation
Intercept	1	1043694	258093	4.04	0.0005	.	0
month_1	1	41693	59018	0.71	0.4873	0.19562	5.11184
month_2	1	23753	48905	0.49	0.6320	0.28489	3.51009
month_3	1	116057	51783	2.24	0.0354	0.25411	3.93535

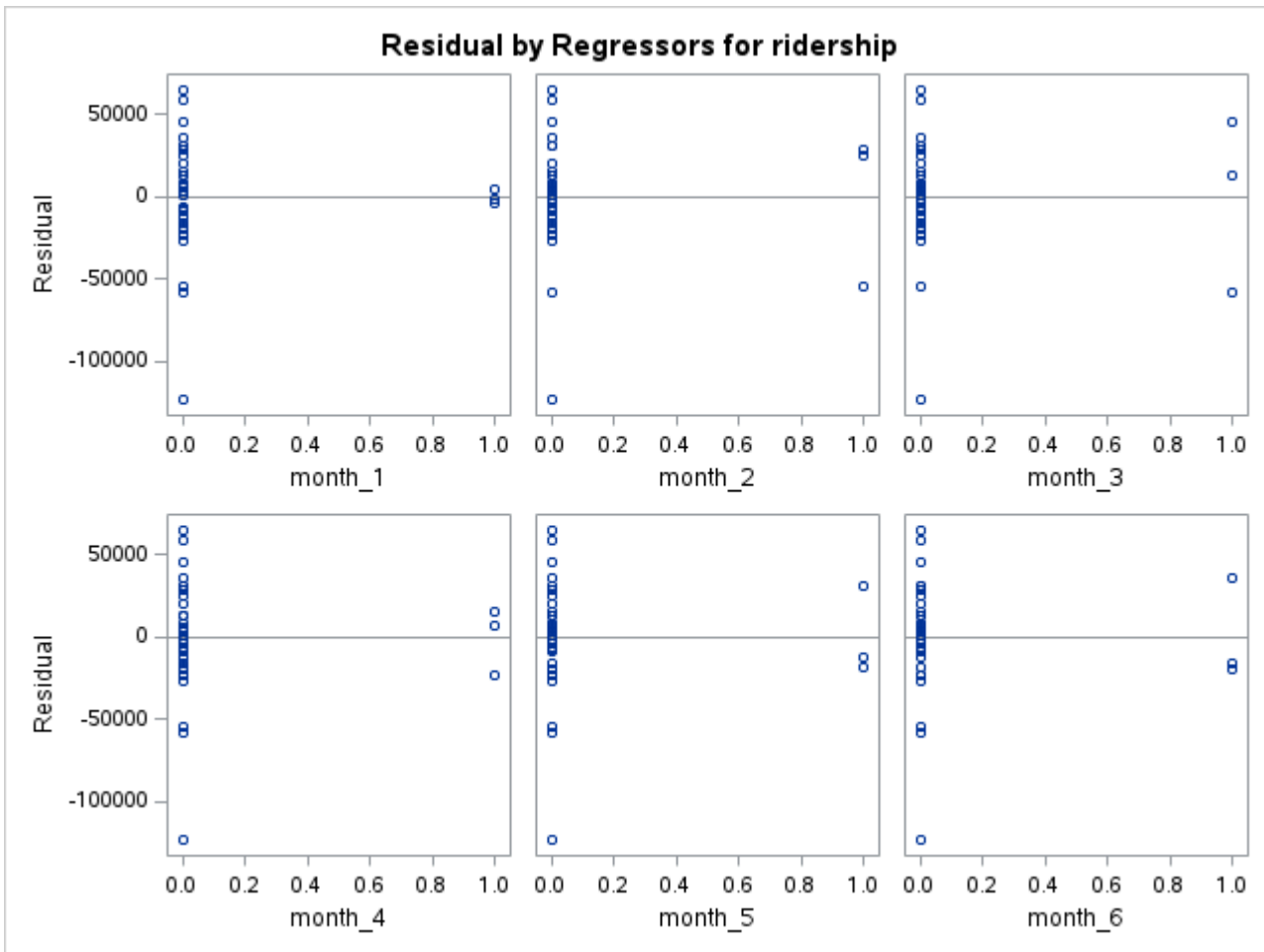
Parameter Estimates

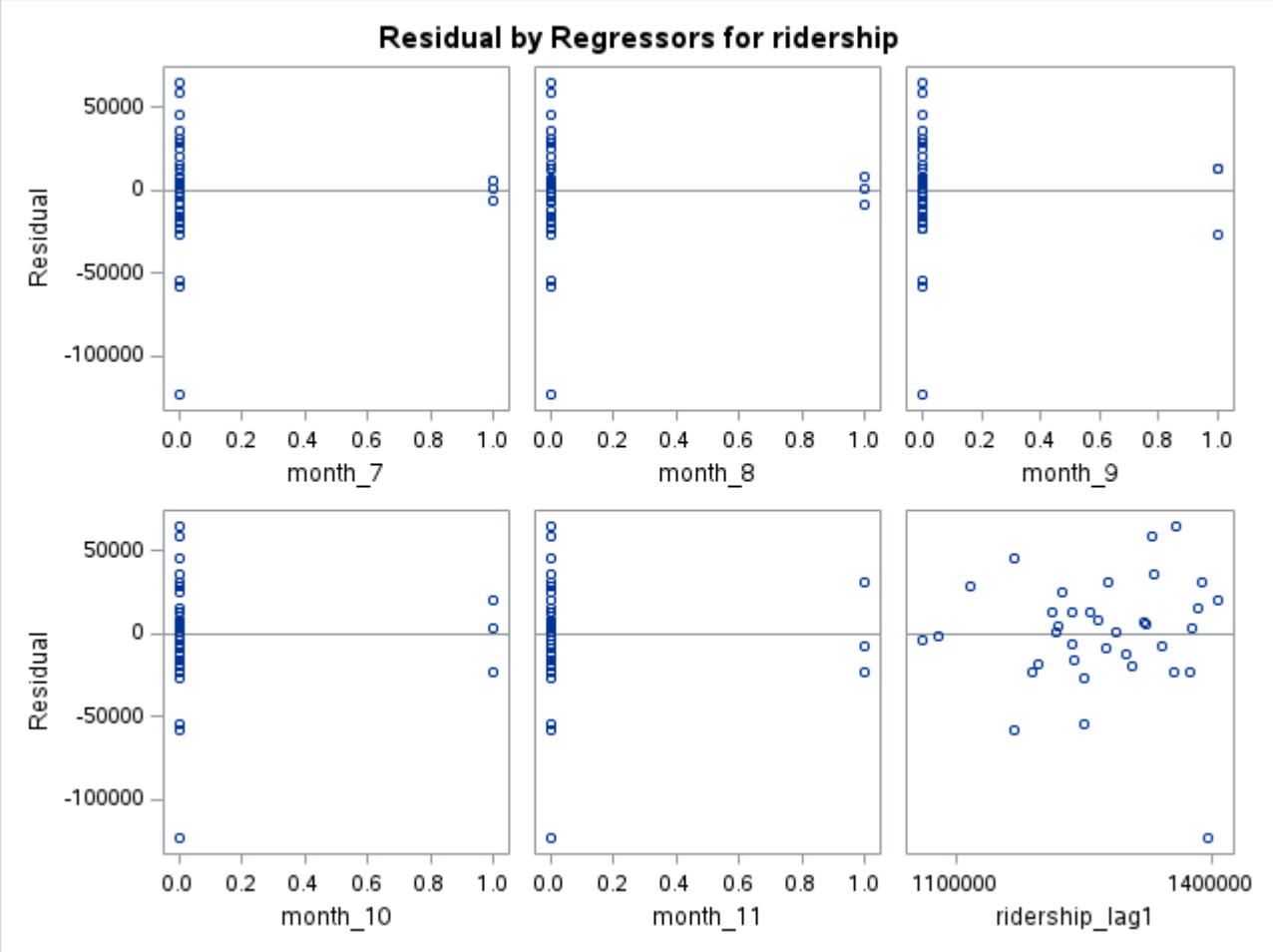
Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > |t|	Tolerance	Variance Inflation
month_4	1	85203	38135	2.23	0.0359	0.46854	2.13431
month_5	1	104217	43751	2.38	0.0263	0.35598	2.80917
month_6	1	95226	39985	2.38	0.0263	0.42619	2.34637
month_7	1	71704	40193	1.78	0.0882	0.42179	2.37085
month_8	1	56622	43655	1.30	0.2081	0.35754	2.79692
month_9	1	166788	51179	3.26	0.0036	0.26014	3.84404
month_10	1	159612	36540	4.37	0.0002	0.51032	1.95955
month_11	1	157772	37193	4.24	0.0003	0.49257	2.03018
ridership_lag1	1	-0.07852	0.21908	-0.36	0.7235	0.14361	6.96351
ridership_lag12	1	0.20554	0.06381	3.22	0.0039	0.38972	2.56595

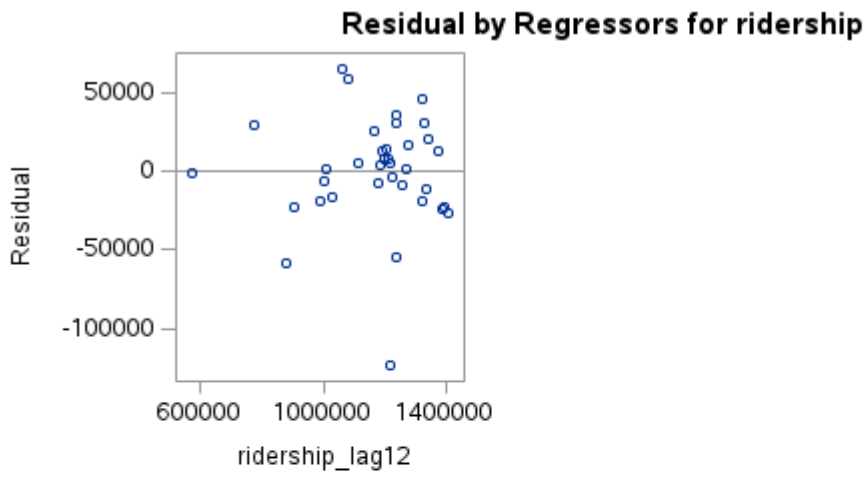
PROC REG Model 2: Lag Baseline

The REG Procedure
Model: MODEL1
Dependent Variable: ridership









Pre-Covid PROC REG Model 3.1

The REG Procedure
Model: MODEL1
Dependent Variable: ridership

Number of Observations Read	36
Number of Observations Used	36

Analysis of Variance

Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	18	2.104088E11	11689378205	7.33	<.0001
Error	17	27110081195	1594710659		
Corrected Total	35	2.375189E11			

Root MSE	39934	R-Square	0.8859
Dependent Mean	1274556	Adj R-Sq	0.7650
Coeff Var	3.13316		

Parameter Estimates

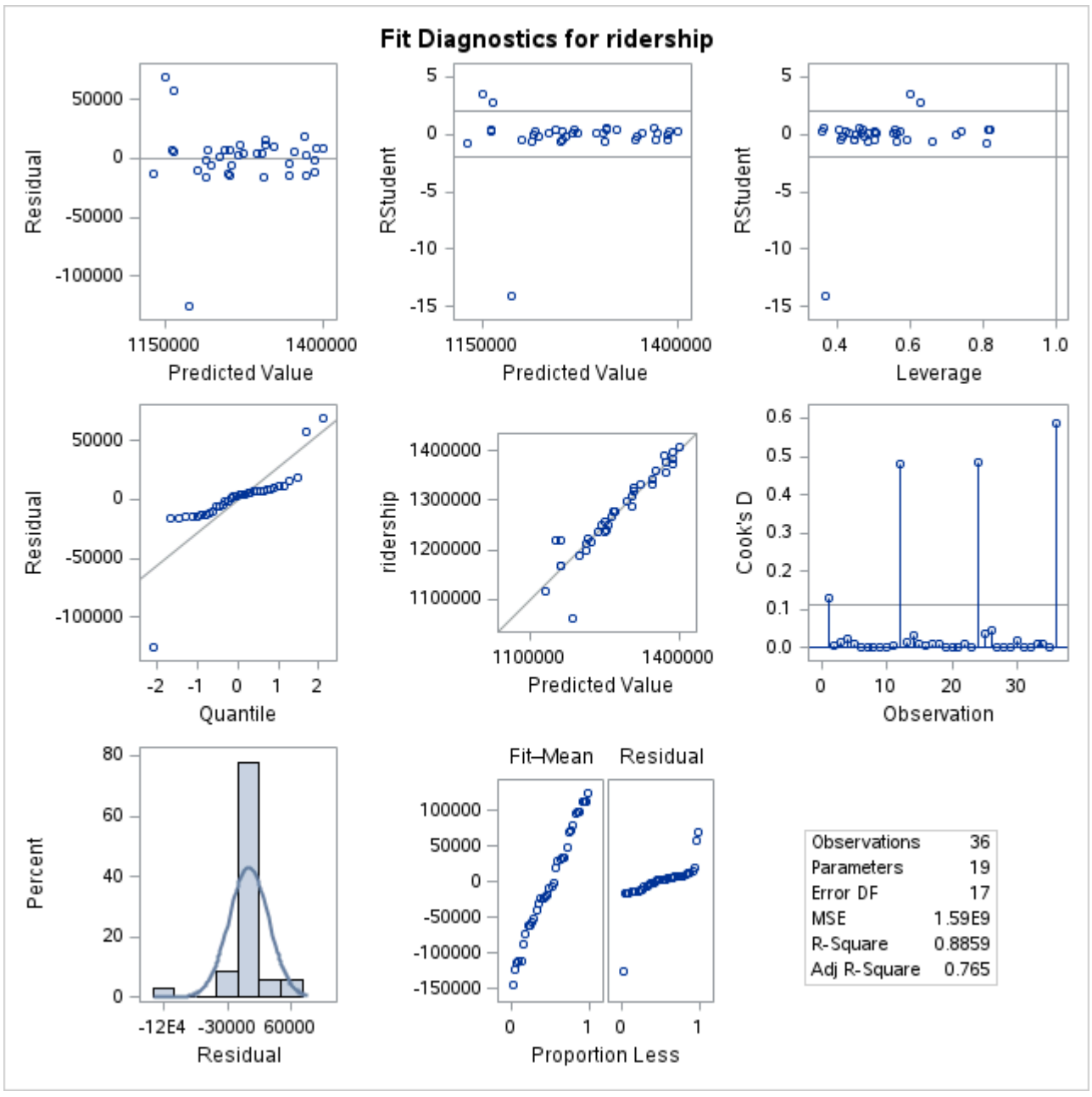
Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > |t|	Tolerance	Variance Inflation
Intercept	1	815514	281541	2.90	0.0100	.	0
month_1	1	82710	58910	1.40	0.1783	0.16710	5.98444
month_2	1	79585	52614	1.51	0.1487	0.20948	4.77371
month_3	1	141799	62474	2.27	0.0365	0.14858	6.73052

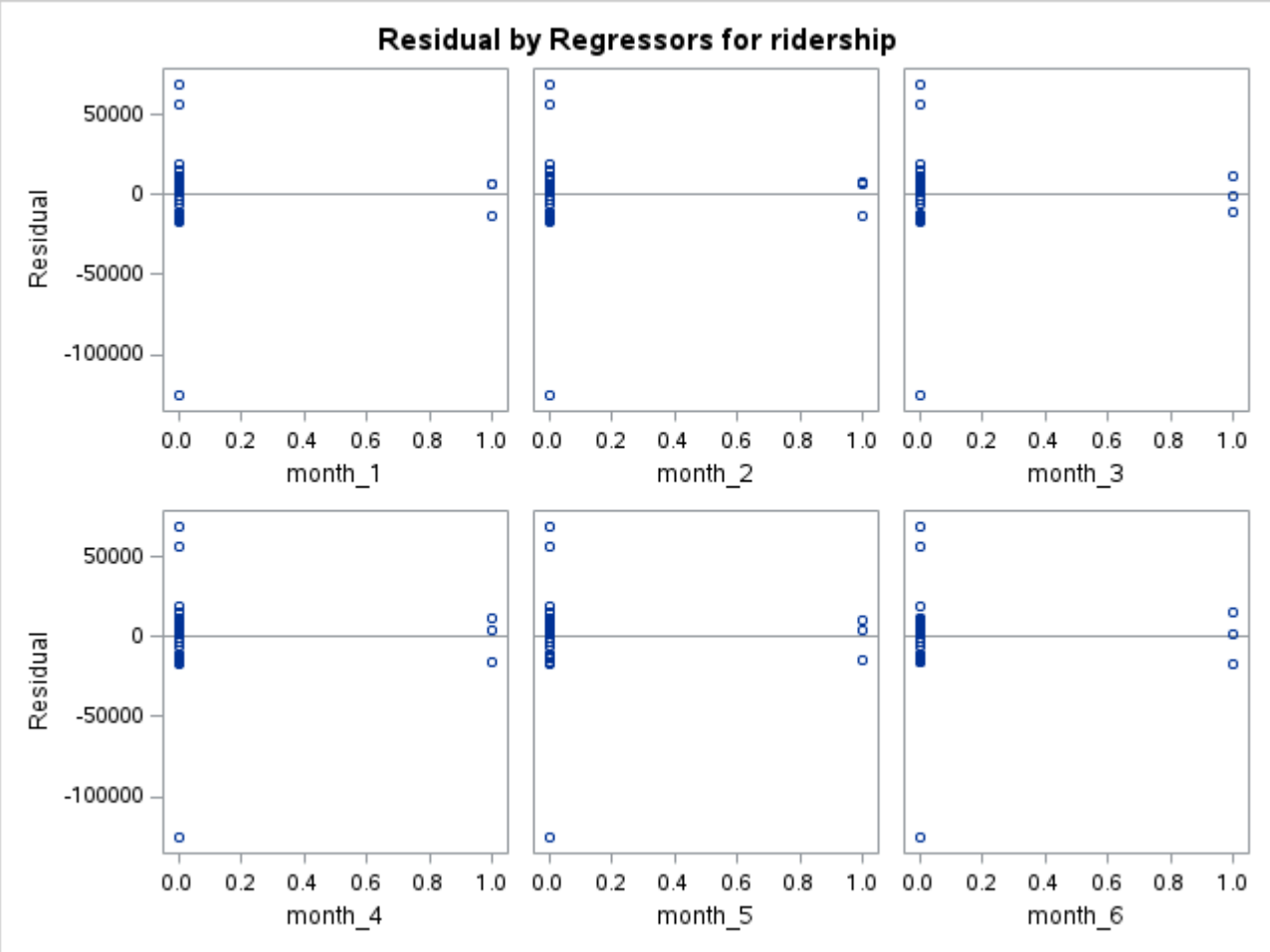
Parameter Estimates

Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > |t|	Tolerance	Variance Inflation
month_4	1	50296	66366	0.76	0.4589	0.13166	7.59528
month_5	1	63515	102868	0.62	0.5451	0.05480	18.24781
month_6	1	41260	131964	0.31	0.7583	0.03330	30.03033
month_7	1	7862.75889	148989	0.05	0.9585	0.02612	38.27903
month_8	1	-4709.31637	139606	-0.03	0.9735	0.02975	33.60921
month_9	1	125634	132174	0.95	0.3552	0.03319	30.12593
month_10	1	115552	79422	1.45	0.1639	0.09193	10.87771
month_11	1	141764	48081	2.95	0.0090	0.25084	3.98656
ridership_lag1	1	0.01121	0.22972	0.05	0.9616	0.11116	8.99569
ridership_lag12	1	0.07224	0.11148	0.65	0.5256	0.10867	9.20193
gas_price_lag1	1	1114.06662	803.95661	1.39	0.1837	0.61373	1.62939
unemployment_rate_lag1	1	17866	16174	1.10	0.2847	0.15206	6.57615
avg_temp	1	1704.74234	6163.17889	0.28	0.7854	0.01387	72.12109
total_precip_mm	1	-133.04720	264.69527	-0.50	0.6217	0.52009	1.92273
total_snow_cm	1	-2327.98886	1105.37436	-2.11	0.0504	0.17811	5.61459

Pre-Covid PROC REG Model 3.1

The REG Procedure
Model: MODEL1
Dependent Variable: ridership




Residual by Regressors for ridership
Residual
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-50000
-100000
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month_1
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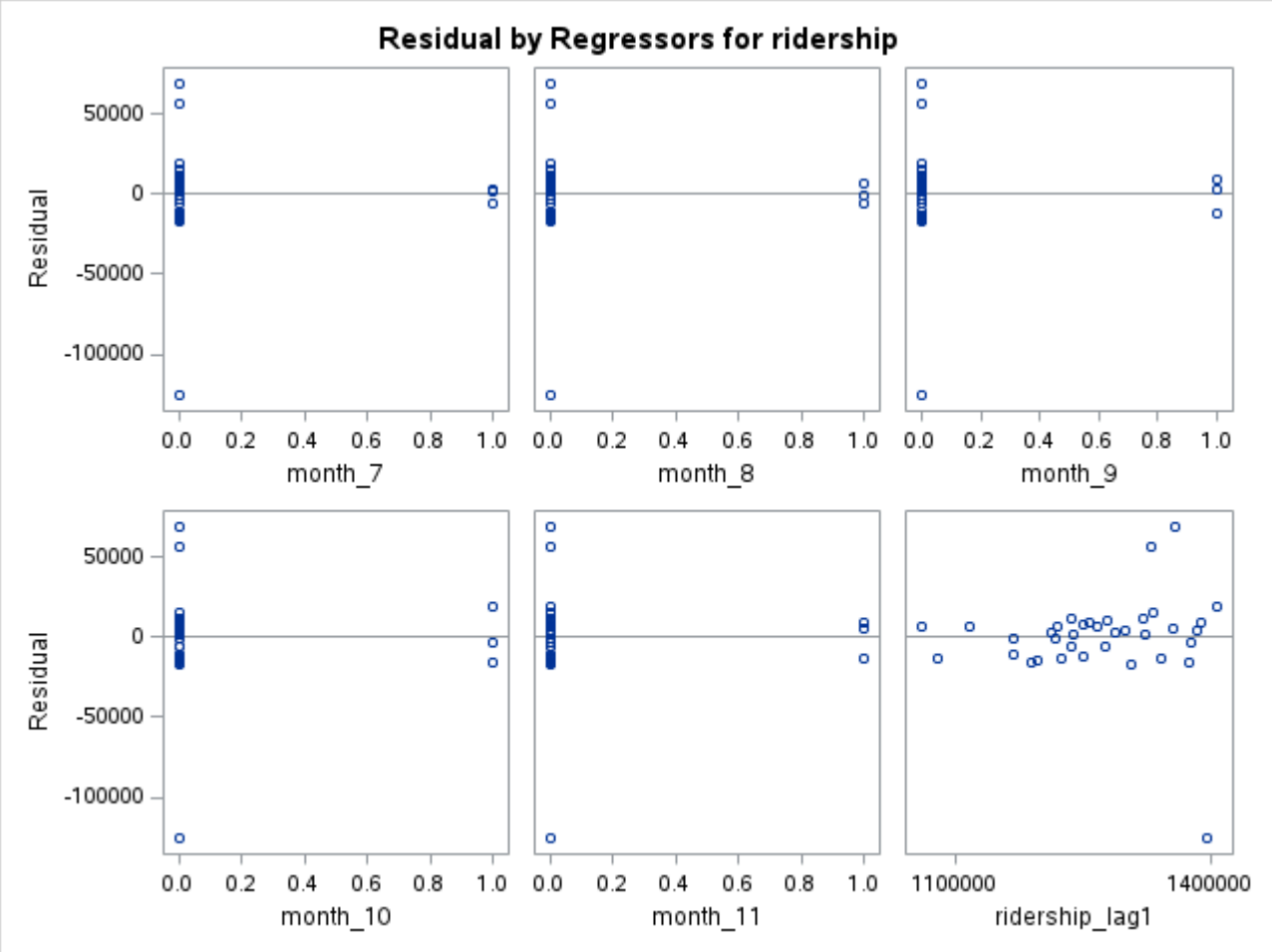
month_4
month_5
month_6


Residual by Regressors for ridership
Residual
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0
-50000
-100000
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month_7
month_8
month_9
month_10
month_11
1100000 1400000
ridership_lag1

Residual by Regressors for ridership
Residual
50000
0
-50000
-100000
600000 1000000 1400000
ridership_lag12
130 140 150 160 170
gas_price_lag1
6 7 8 9
unemployment_rate_lag1
0 10 20
avg_temp
50 100 150
total_precip_mm
0 20 40 60
total_snow_cm

Pre-Covid PROC REG STEP Model 3.2

The REG Procedure
Model: MODEL1
Dependent Variable: ridership

Number of Observations Read	36
Number of Observations Used	36

Stepwise Selection: Step 1

Variable ridership_lag12 Entered: R-Square = 0.4970 and C(p) = 42.9125

Analysis of Variance					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	1	1.180551E11	1.180551E11	33.60	<.0001
Error	34	1.194638E11	3513642313		
Corrected Total	35	2.375189E11			

Variable	Parameter Estimate	Standard Error	Type II SS	F Value	Pr > F
Intercept	904211	64651	6.87308E11	195.61	<.0001
ridership_lag12	0.31619	0.05455	1.180551E11	33.60	<.0001

Bounds on condition number: 1, 1

Stepwise Selection: Step 2

Variable unemployment_rate_lag1 Entered: R-Square = 0.5936 and C(p) = 30.5235

Analysis of Variance					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	2	1.410015E11	70500726116	24.10	<.0001
Error	33	96517436658	2924770808		
Corrected Total	35	2.375189E11			

Variable	Parameter Estimate	Standard Error	Type II SS	F Value	Pr > F
Intercept	988064	66147	6.525928E11	223.13	<.0001
ridership_lag12	0.48813	0.07903	1.115899E11	38.15	<.0001
unemployment_rate_lag1	-37990	13563	22946401983	7.85	0.0085

Bounds on condition number: 2.5214, 10.085

All variables left in the model are significant at the 0.1500 level.

No other variable met the 0.0500 significance level for entry into the model.

Summary of Stepwise Selection								
Step	Variable Entered	Variable Removed	Number Vars In	Partial R-Square	Model R-Square	C(p)	F Value	Pr > F
1	ridership_lag12		1	0.4970	0.4970	42.9125	33.60	<.0001
2	unemployment_rate_lag1		2	0.0966	0.5936	30.5235	7.85	0.0085

Pre-Covid PROC REG STEP Model 3.2

The REG Procedure
Model: MODEL1
Dependent Variable: ridership

Number of Observations Read	36
Number of Observations Used	36

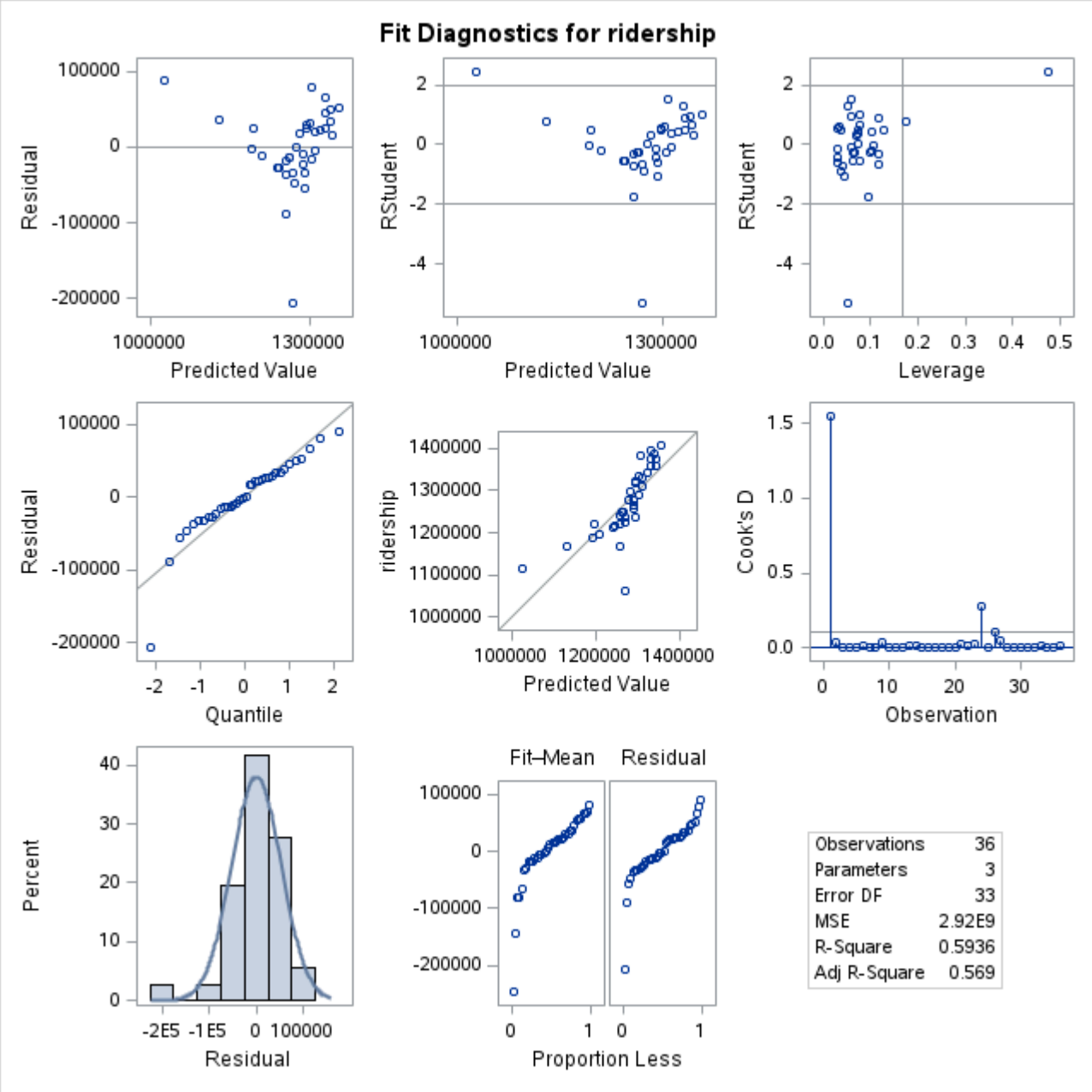
Analysis of Variance					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	2	1.410015E11	70500726116	24.10	<.0001
Error	33	96517436658	2924770808		
Corrected Total	35	2.375189E11			

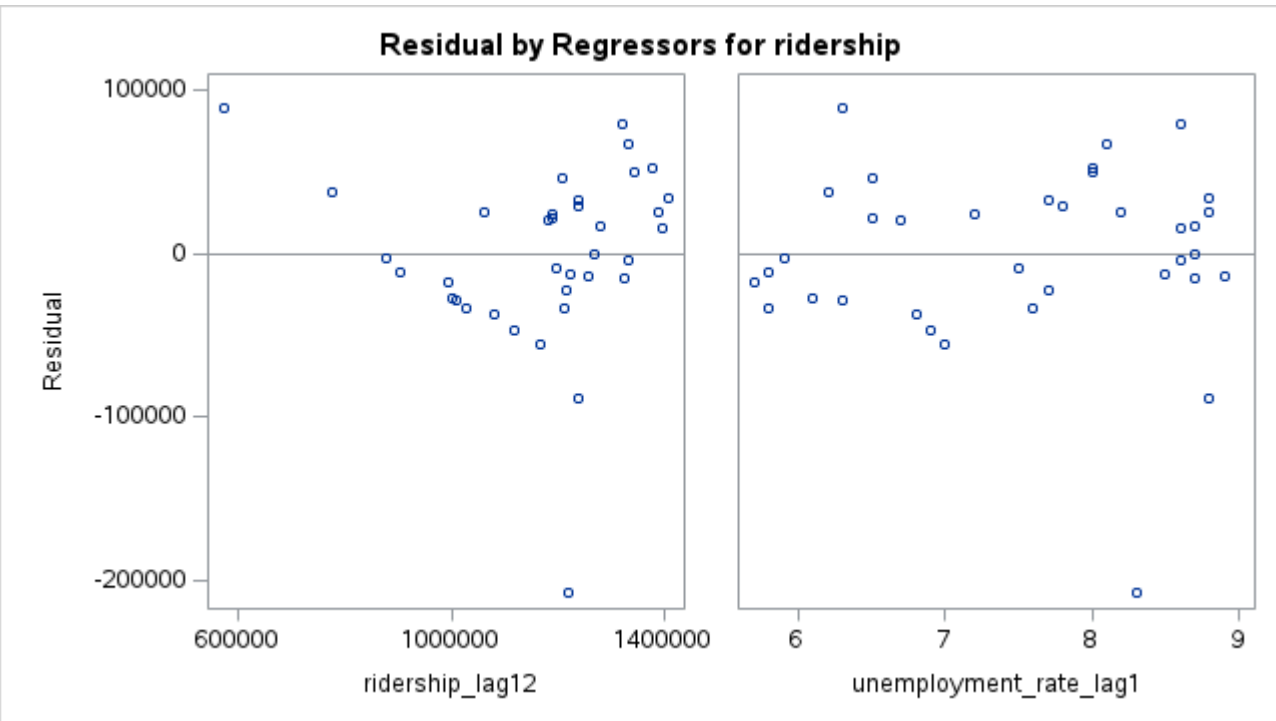
Root MSE	54081	R-Square	0.5936
Dependent Mean	1274556	Adj R-Sq	0.5690
Coeff Var	4.24314		

Parameter Estimates							
Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > |t|	Tolerance	Variance Inflation
Intercept	1	988064	66147	14.94	<.0001	.	0
ridership_lag12	1	0.48813	0.07903	6.18	<.0001	0.39661	2.52137
unemployment_rate_lag1	1	-37990	13563	-2.80	0.0085	0.39661	2.52137

Pre-Covid PROC REG STEP Model 3.2

The REG Procedure
Model: MODEL1
Dependent Variable: ridership





Pre-COVID Model P1: Month-only Seasonal Baseline

The REG Procedure
Model: MODEL1
Dependent Variable: ridership

Number of Observations Read	36
Number of Observations Used	36

Analysis of Variance

Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	11	1.664229E11	15129353535	5.11	0.0004
Error	24	71096000000	2962333333		
Corrected Total	35	2.375189E11			

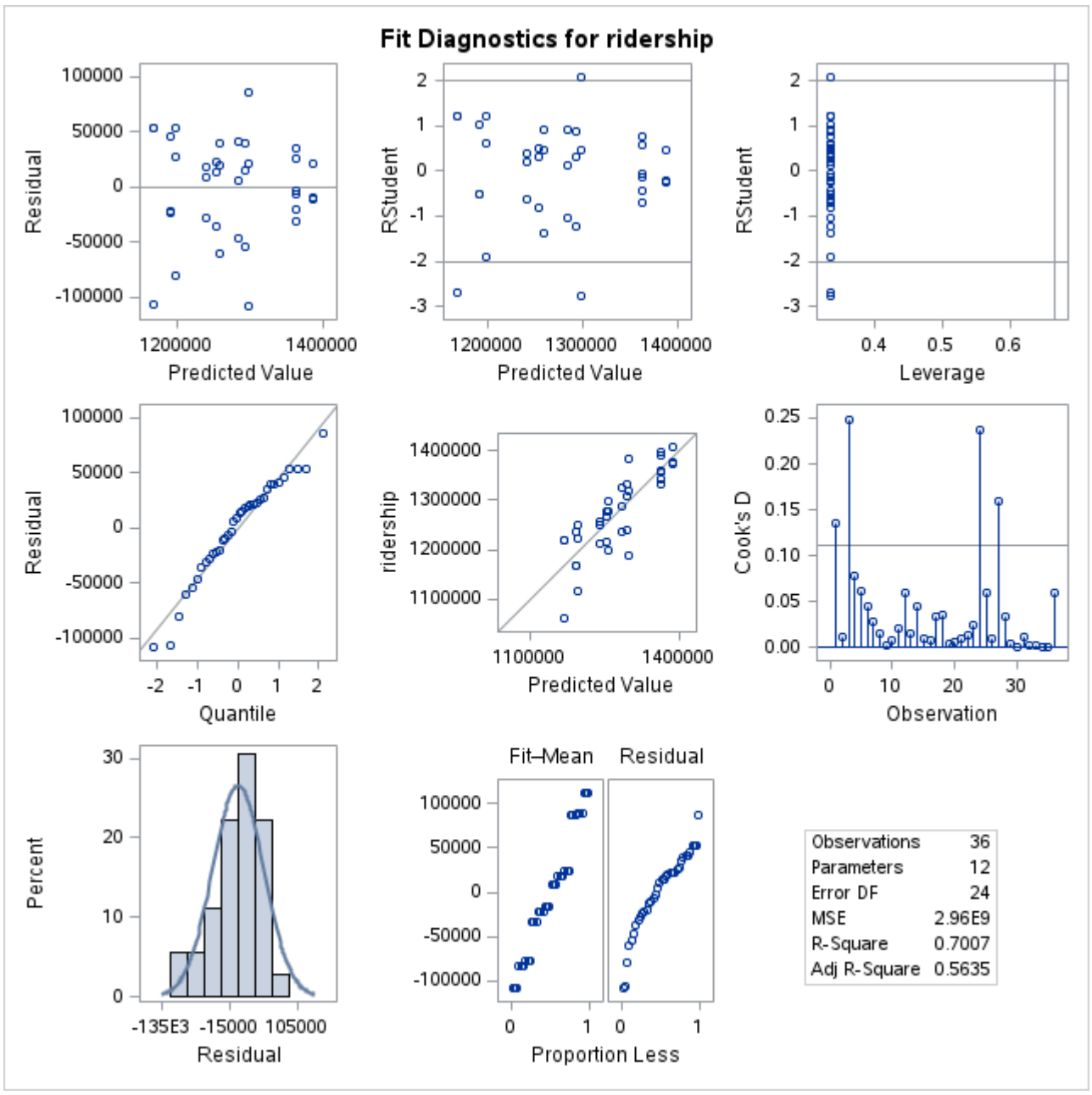
Root MSE	54427	**R-Square**	0.7007
Dependent Mean	1274556	**Adj R-Sq**	0.5635
Coeff Var	4.27030		

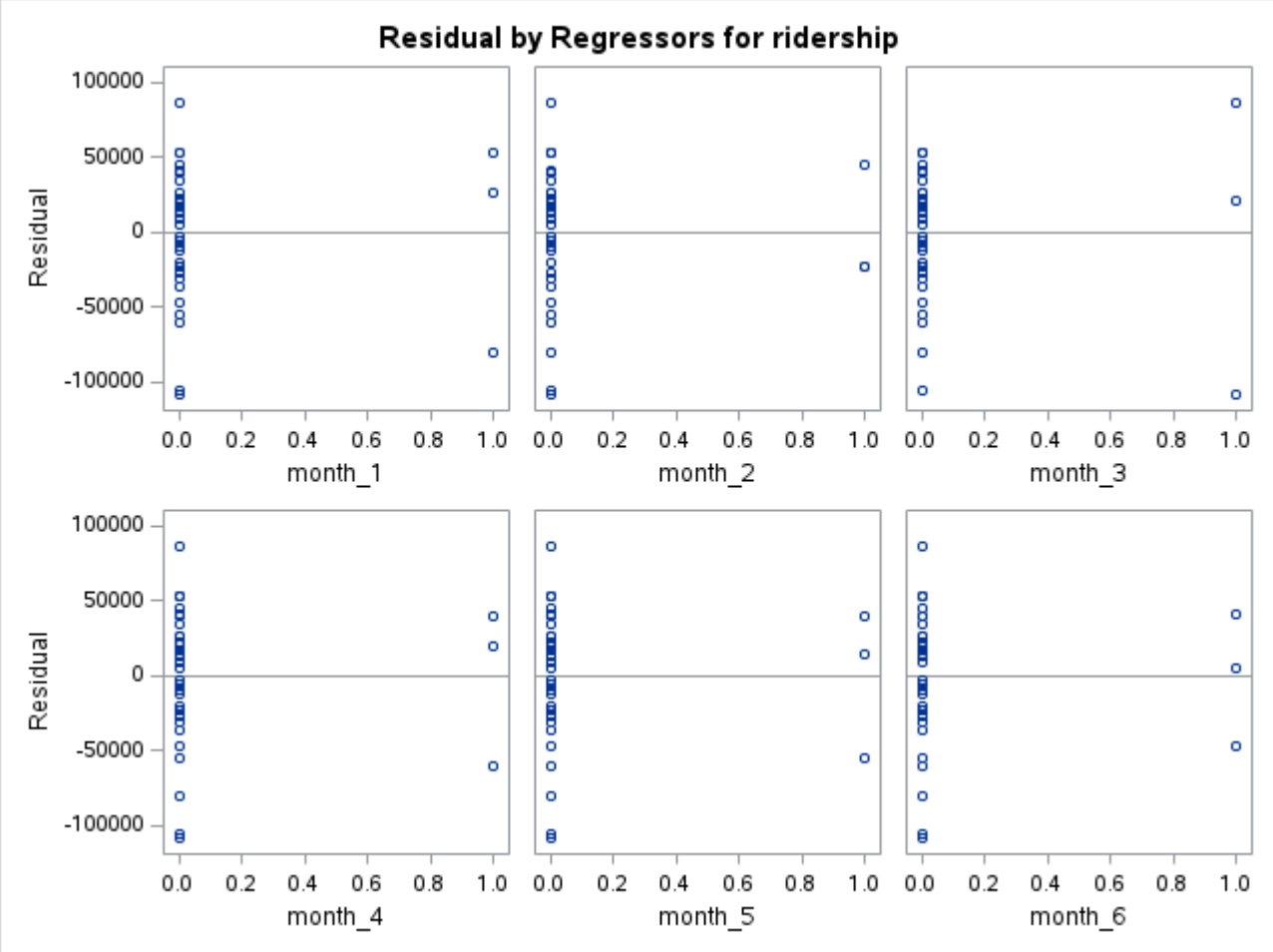
Parameter Estimates

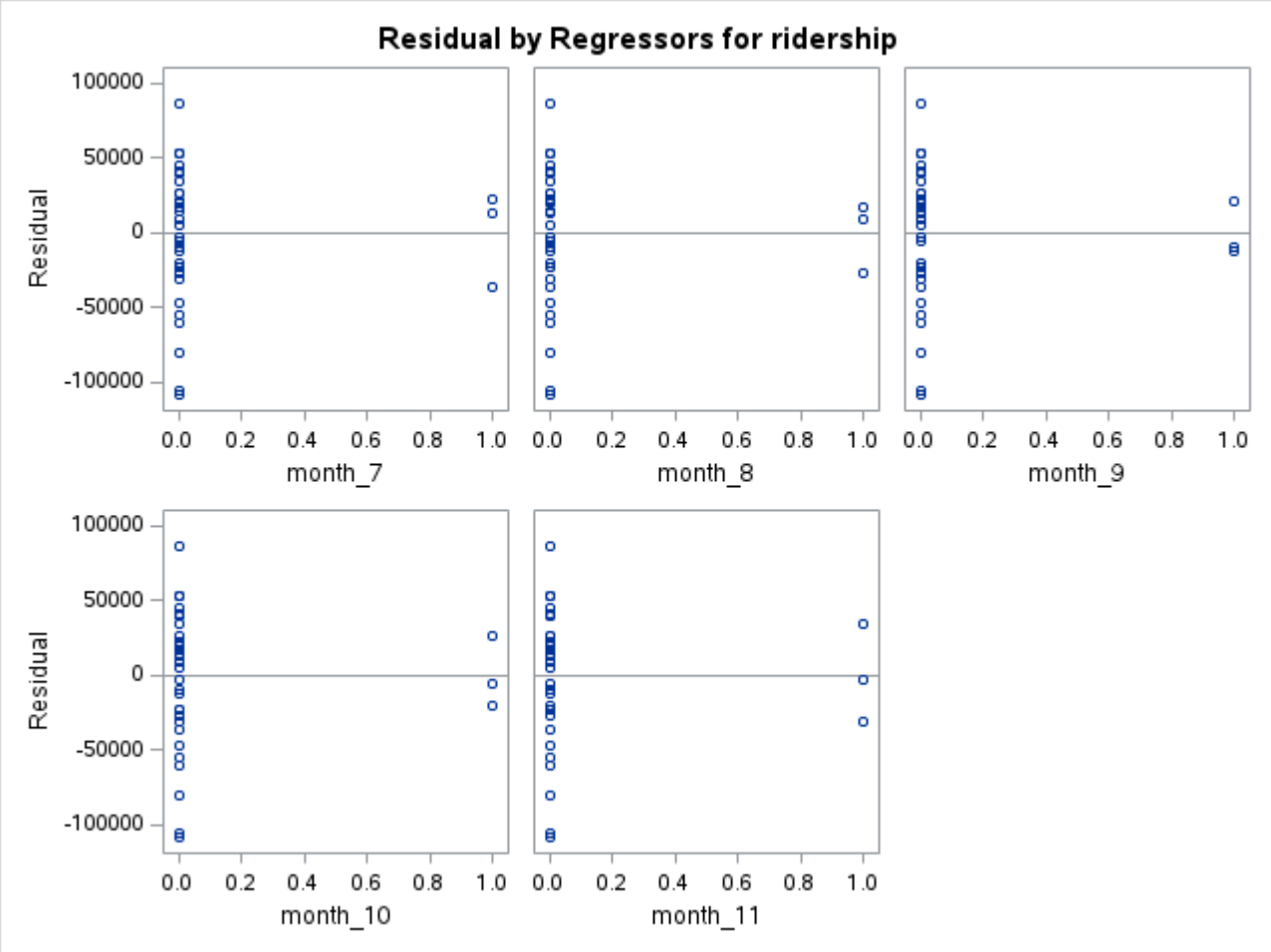
Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > |t|	Tolerance	Variance Inflation
Intercept	1	1167000	31424	37.14	<.0001	.	0
month_1	1	30000	44440	0.68	0.5061	0.54545	1.83333
month_2	1	24333	44440	0.55	0.5891	0.54545	1.83333
month_3	1	131333	44440	2.96	0.0069	0.54545	1.83333
month_4	1	91667	44440	2.06	0.0501	0.54545	1.83333
month_5	1	126333	44440	2.84	0.0090	0.54545	1.83333
month_6	1	116333	44440	2.62	0.0151	0.54545	1.83333
month_7	1	86333	44440	1.94	0.0639	0.54545	1.83333
month_8	1	73333	44440	1.65	0.1119	0.54545	1.83333
month_9	1	219667	44440	4.94	<.0001	0.54545	1.83333
month_10	1	196000	44440	4.41	0.0002	0.54545	1.83333
month_11	1	195333	44440	4.40	0.0002	0.54545	1.83333

Pre-COVID Model P1: Month-only Seasonal Baseline

The REG Procedure
Model: MODEL1
Dependent Variable: ridership







Pre-COVID Model P2: Lag Memory Model

The REG Procedure
Model: MODEL1
Dependent Variable: ridership

Number of Observations Read	36
Number of Observations Used	36

Analysis of Variance

Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	2	1.195847E11	59792366115	16.73	<.0001
Error	33	1.179342E11	3573762323		
Corrected Total	35	2.375189E11			

Root MSE	59781	**R-Square**	0.5035
Dependent Mean	1274556	**Adj R-Sq**	0.4734
Coeff Var	4.69034		

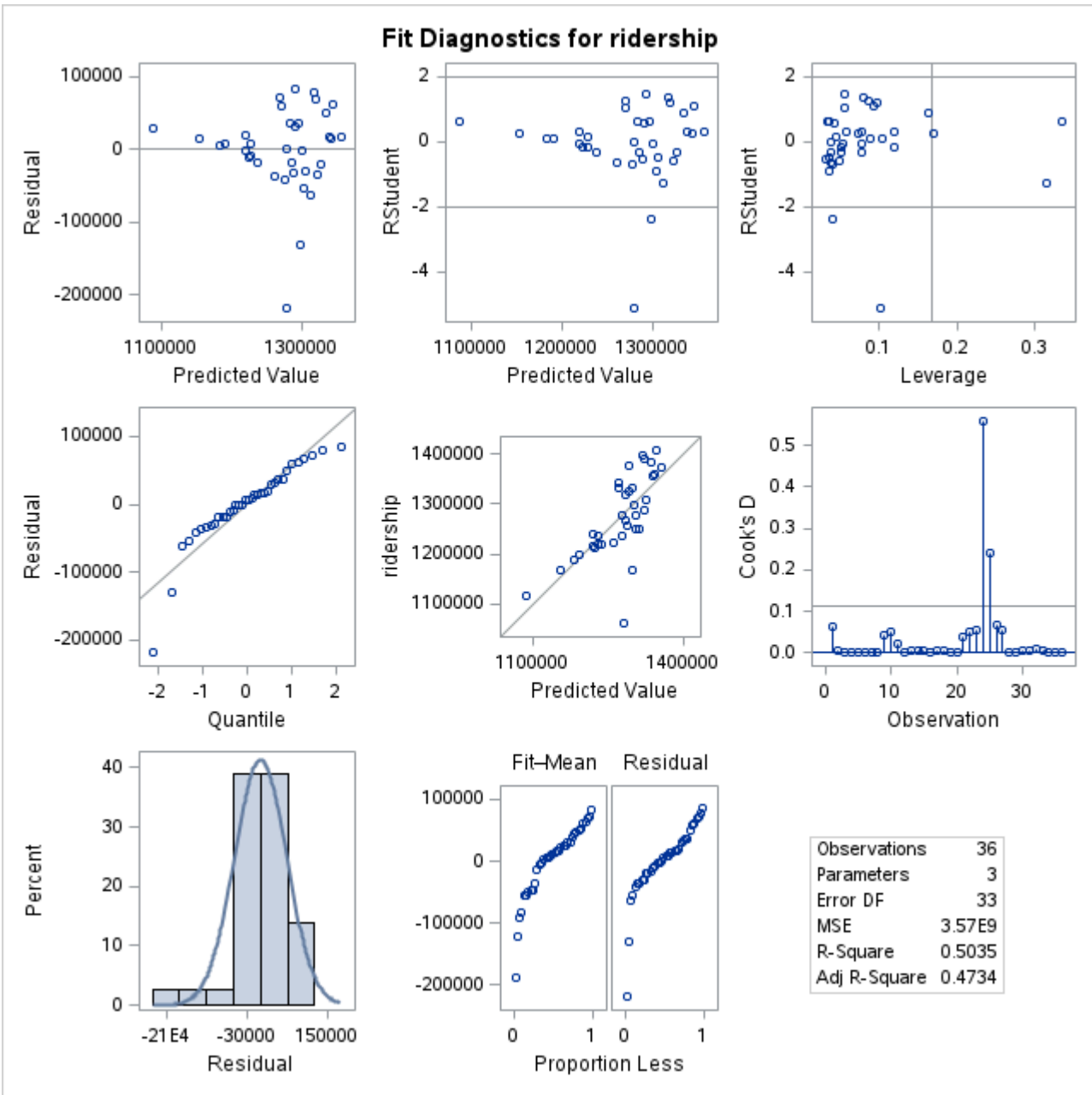
Parameter Estimates

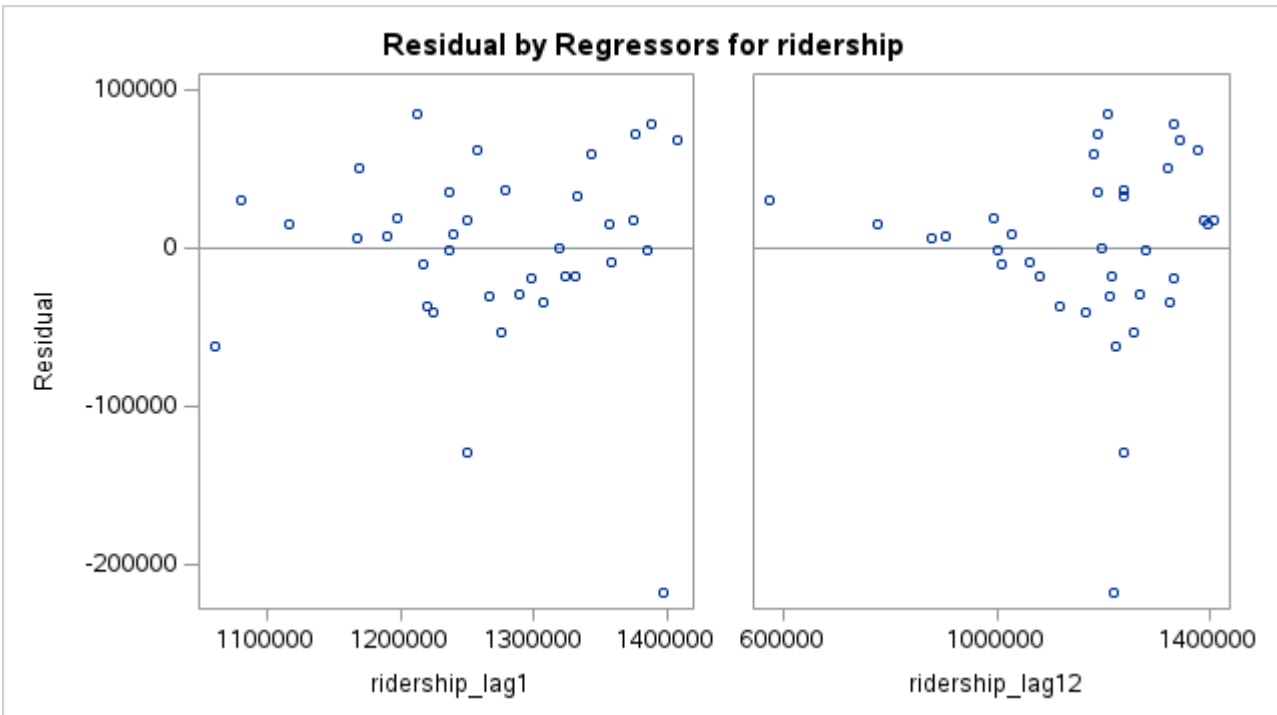
Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > |t|	Tolerance	Variance Inflation
Intercept	1	991495	148492	6.68	<.0001	.	0
ridership_lag1	1	-0.09302	0.14218	-0.65	0.5175	0.65033	1.53769

Parameter Estimates							
Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > |t|	Tolerance	Variance Inflation
ridership_lag12	1	0.34258	0.06822	5.02	<.0001	0.65033	1.53769

Pre-COVID Model P2: Lag Memory Model

The REG Procedure
Model: MODEL1
Dependent Variable: ridership





Pre-COVID Model P3: Reduced External Model

The REG Procedure
Model: MODEL1
Dependent Variable: ridership

Number of Observations Read	36
Number of Observations Used	36

Analysis of Variance					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	4	1.439579E11	35989487302	11.92	<.0001

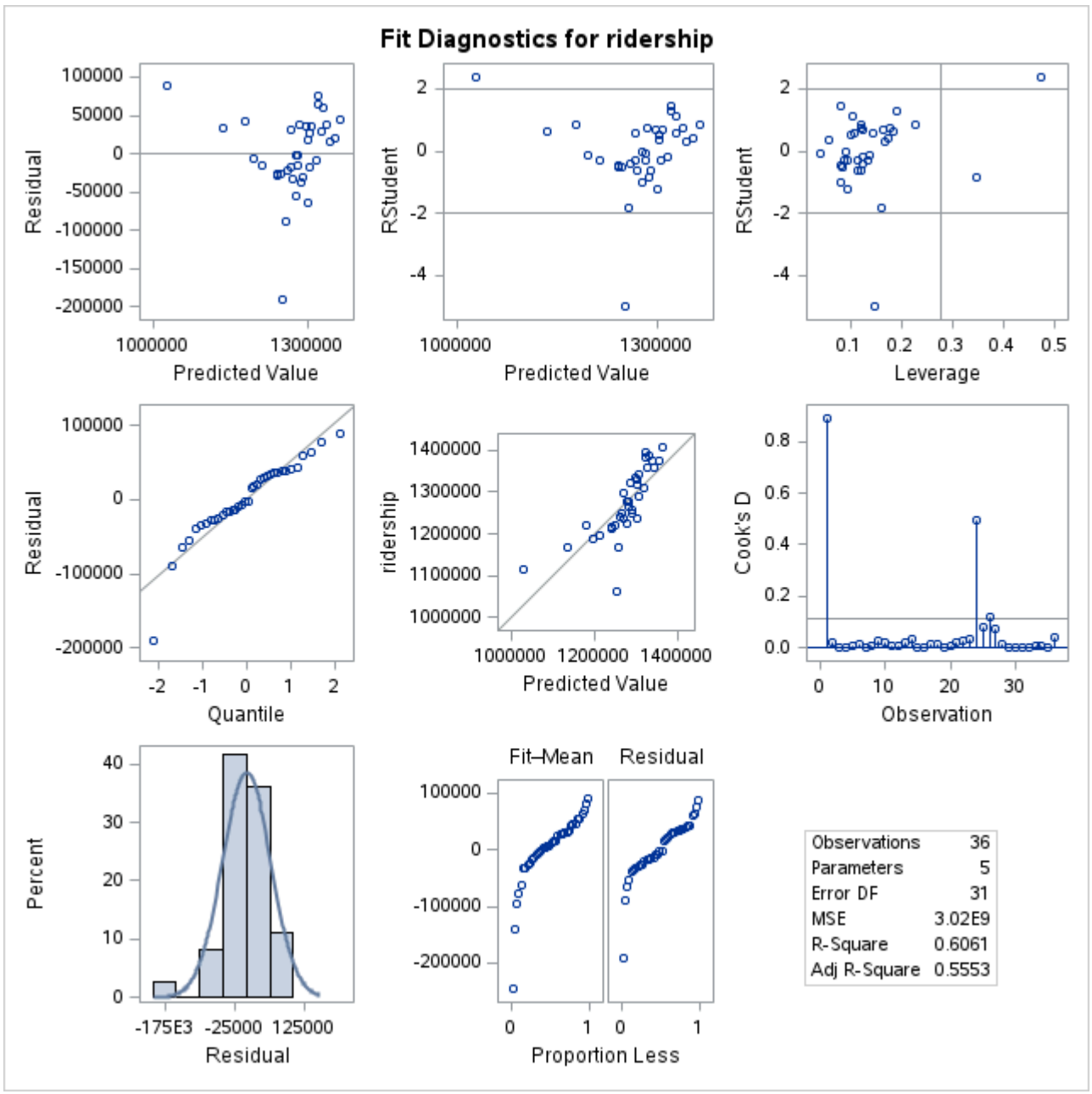
Analysis of Variance					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Error	31	93560939683	3018094828		
Corrected Total	35	2.375189E11			

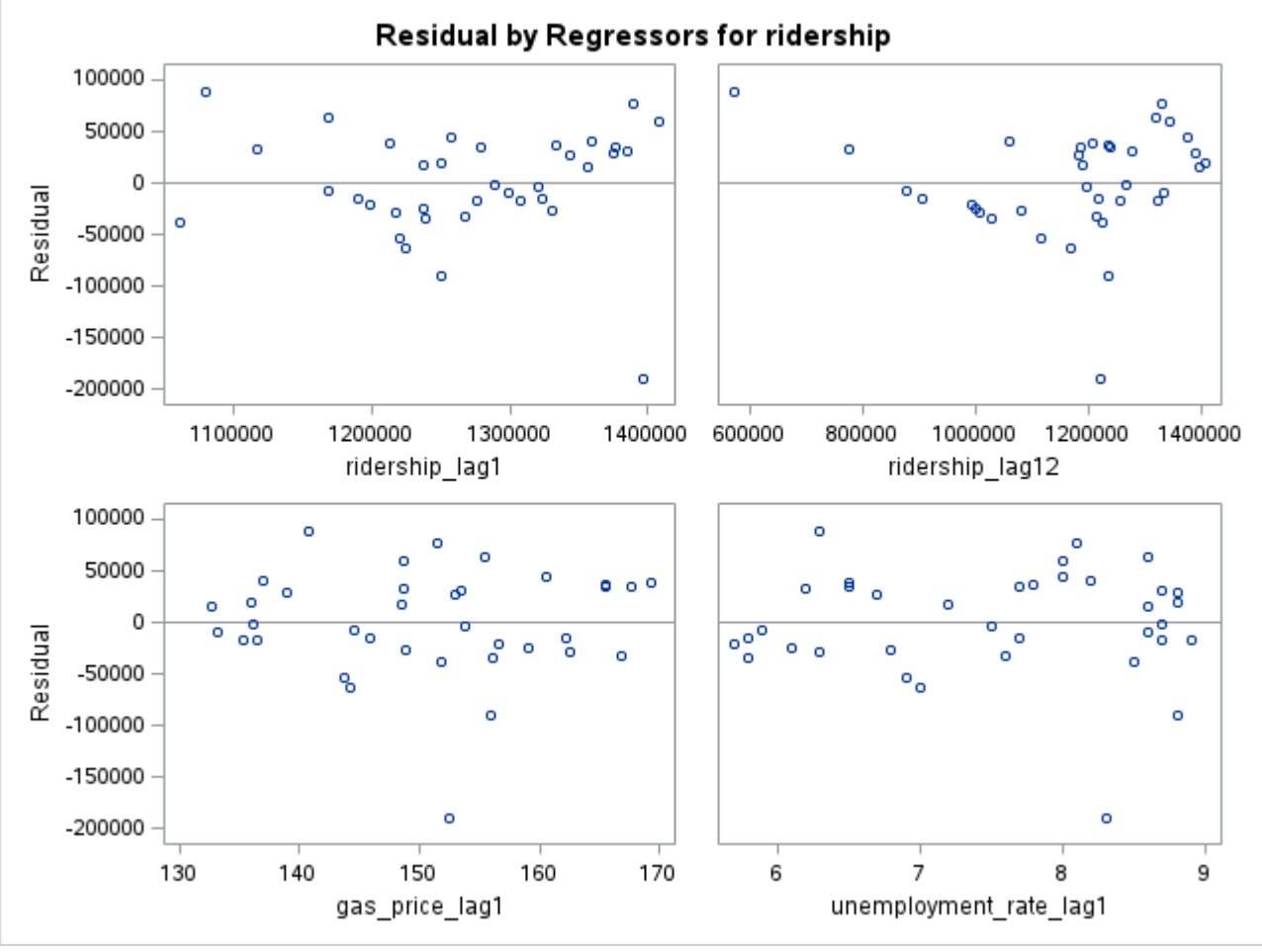
Root MSE	54937	R-Square	0.6061
Dependent Mean	1274556	Adj R-Sq	0.5553
Coeff Var	4.31030		

Parameter Estimates							
Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > |t|	Tolerance	Variance Inflation
Intercept	1	1150372	229458	5.01	<.0001	.	0
ridership_lag1	1	-0.12783	0.13135	-0.97	0.3380	0.64348	1.55405
ridership_lag12	1	0.53766	0.09792	5.49	<.0001	0.26656	3.75151
gas_price_lag1	1	-231.08049	1009.47814	-0.23	0.8204	0.73671	1.35738
unemployment_rate_lag1	1	-41068	16097	-2.55	0.0159	0.29056	3.44159

Pre-COVID Model P3: Reduced External Model

The REG Procedure
Model: MODEL1
Dependent Variable: ridership





Pre-Covid Model P4: Whithout Ridership History

The REG Procedure
Model: MODEL1
Dependent Variable: ridership

Number of Observations Read	36
Number of Observations Used	36

Analysis of Variance

Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	5	1.24775E11	24954996154	6.64	0.0003
Error	30	1.127439E11	3758130271		
Corrected Total	35	2.375189E11			

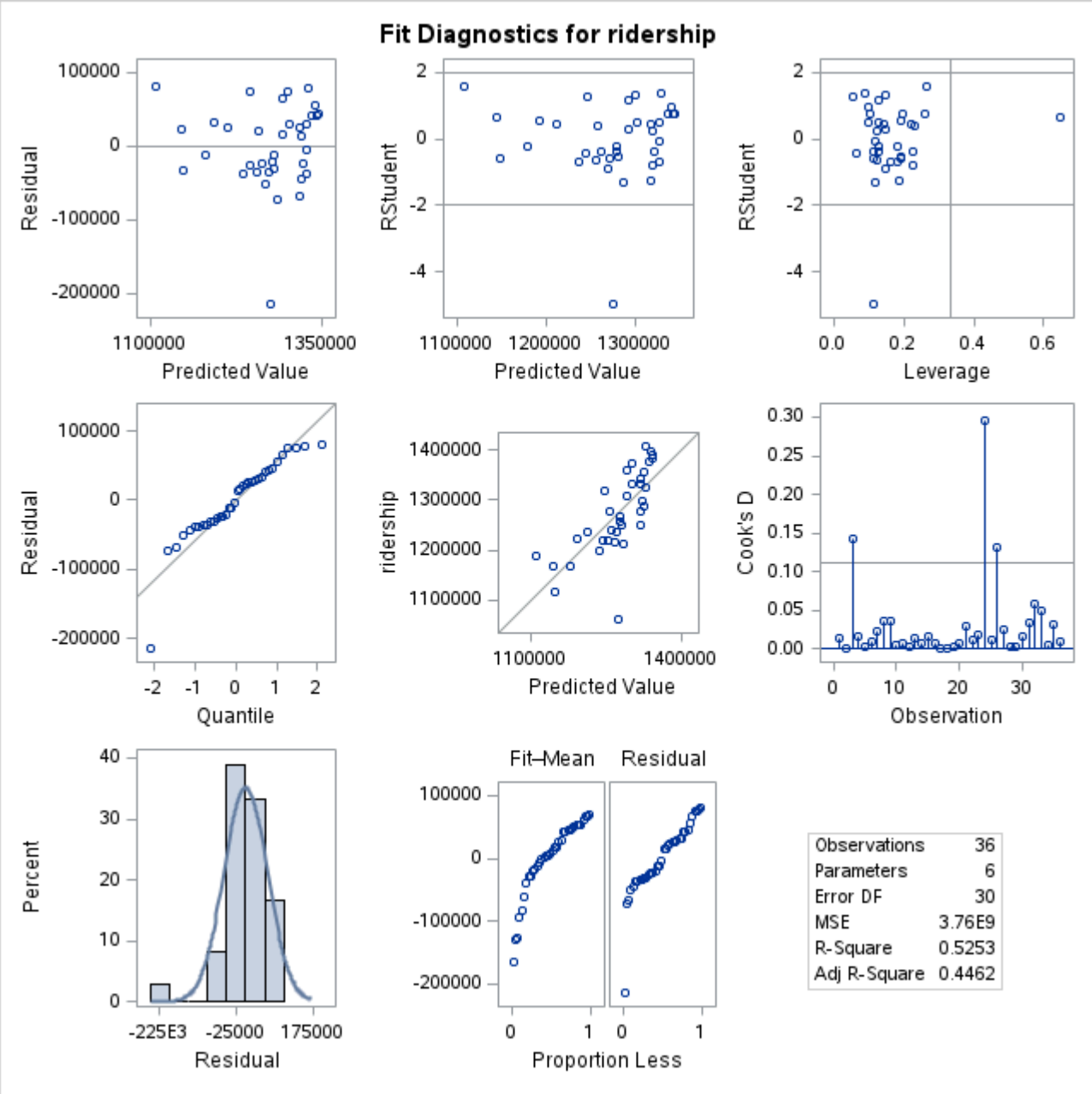
Root MSE	61304	R-Square	0.5253
Dependent Mean	1274556	Adj R-Sq	0.4462
Coeff Var	4.80980		

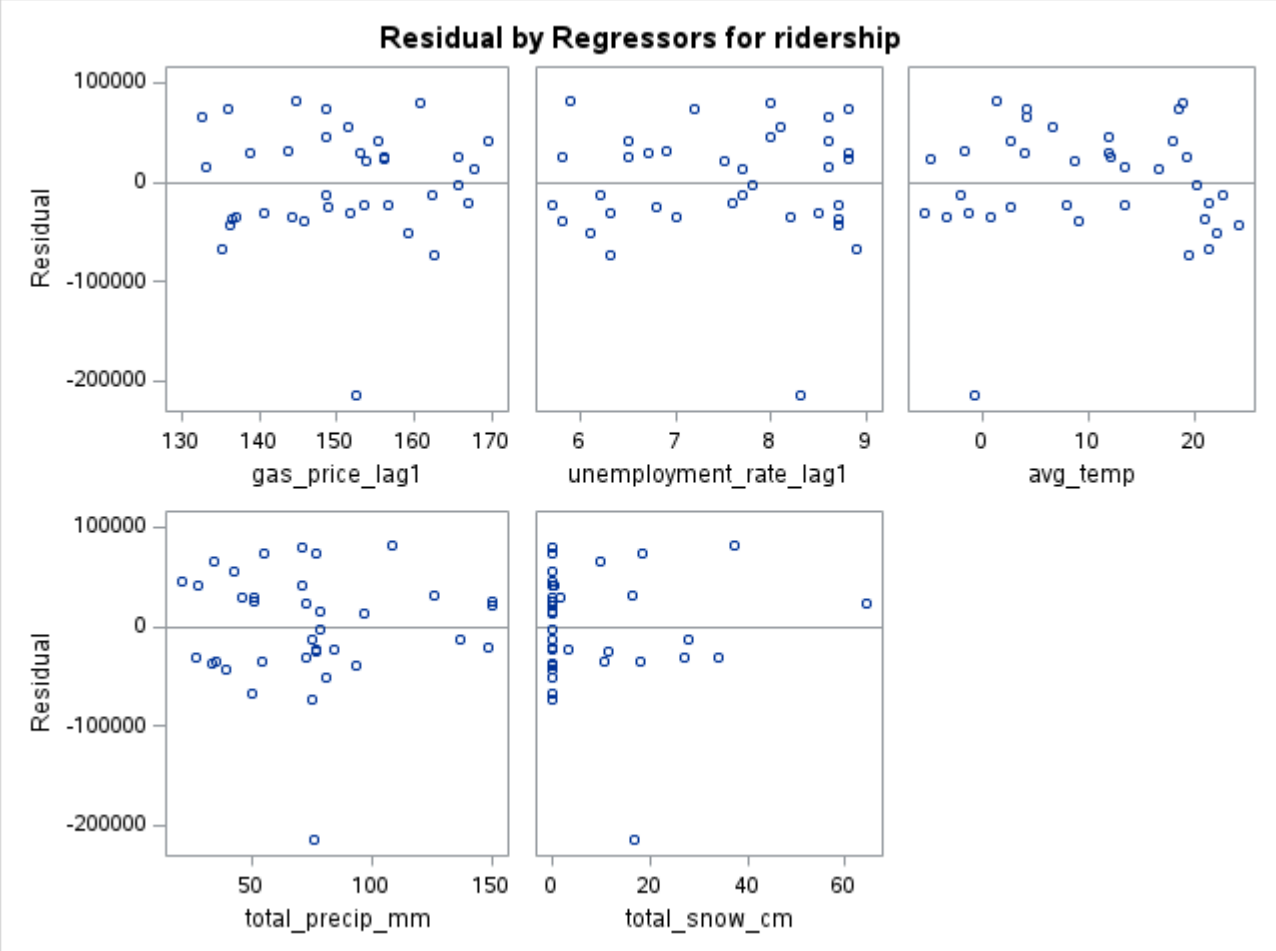
Parameter Estimates

Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > |t|	Tolerance	Variance Inflation
Intercept	1	882329	203872	4.33	0.0002	.	0
gas_price_lag1	1	1852.73421	1098.41748	1.69	0.1020	0.77481	1.29064
unemployment_rate_lag1	1	25816	10814	2.39	0.0235	0.80159	1.24752
avg_temp	1	-582.58563	1682.64491	-0.35	0.7316	0.43838	2.28112
total_precip_mm	1	-653.05415	321.34629	-2.03	0.0511	0.83161	1.20249
total_snow_cm	1	-3248.84265	1045.26520	-3.11	0.0041	0.46939	2.13040

Pre-Covid Model P4: Whithout Ridership History

The REG Procedure
Model: MODEL1
Dependent Variable: ridership


Fit Diagnostics for ridership
Residual
Predicted Value
RStudent
Leverage
Quantile
ridership
Cook's D
Observation
Percent
Fit–Mean
Proportion Less
Observations 36
Parameters 6
Error DF 30
MSE 3.76E9
R-Square 0.5253
Adj R-Square 0.4462


Residual by Regressors for ridership
Residual
gas_price_lag1
unemployment_rate_lag1
avg_temp
total_precip_mm
total_snow_cm